

Fentanyl: Drug information - UpToDate

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For additional information see "Fentanyl: Patient drug information" and "Fentanyl: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Special Alerts

Opioid Labeling Safety Updates August 2025

The FDA is requiring safety labeling changes for all opioid pain medications to better highlight the risks of long-term use, including addiction, misuse, and overdose. This decision follows a public advisory meeting that reviewed data from 2 large postmarketing studies, which revealed serious risks and a lack of evidence supporting long-term opioid effectiveness. The updated labels will include clearer risk information, stronger dosing warnings, revised guidance on appropriate use, details on overdose reversal agents, and drug interactions. They will also stress safe discontinuation to prevent harm, outline new risks like overdose-related brain injury and digestive issues, and strengthen warnings about interactions with CNS depressants, including gabapentinoids. Additionally, the FDA is requiring a new clinical trial to further assess long-term opioid use.

Further information may be found at: <https://www.fda.gov/drugs/drug-safety-and-availability/fda-requiring-opioid-pain-medicine-manufacturers-update-prescribing-information-regarding-long-term>

ALERT: US Boxed Warning

Life-threatening respiratory depression:

Buccal, intranasal, lozenge, sublingual: Serious, life-threatening, or fatal respiratory depression has occurred with use of fentanyl, including following use in opioid non-tolerant patients and improper dosing. Evaluate patients for respiratory depression, especially during initiation of fentanyl or following a dosage increase. To reduce the risk of respiratory depression, proper dosing and titration of fentanyl are essential. The substitution of fentanyl buccal, intranasal, lozenge, or sublingual for any other fentanyl product may result in fatal overdose. Due to the risk of respiratory depression, fentanyl buccal, intranasal, lozenge, and sublingual are contraindicated in the management of acute or postoperative pain including headache/migraine and in opioid non-tolerant patients.

Parenteral, transdermal patch: Serious, life-threatening, or fatal respiratory depression may occur with use of fentanyl, especially during initiation or following a dosage increase. To reduce the risk of respiratory depression, proper dosing and titration of fentanyl parenteral and transdermal system are essential.

Addiction, abuse, and misuse:

Because the use of fentanyl exposes patients and other users to the risks of opioid addiction, abuse, and misuse, which can lead to overdose and death, assess each patient's risk prior to prescribing and reassess all patients regularly for the development of these behaviors and conditions.

Accidental exposure:

Accidental exposure of even one dose of fentanyl, especially by children, can result in a fatal overdose of fentanyl. Death has been reported in children who have accidentally ingested transmucosal immediate-release fentanyl products. Deaths due to an overdose of fentanyl have occurred when children and adults were accidentally exposed to fentanyl transdermal system. Strict adherence to the recommended handling and disposal instructions is of the utmost importance to prevent accidental exposure. Fentanyl must be kept out of reach of children.

Cytochrome P450 3A4 interaction:

The concomitant use of fentanyl with all cytochrome P450 3A4 inhibitors may result in an increase in fentanyl plasma concentrations, which could increase or prolong adverse reactions and may cause potentially fatal respiratory depression. In addition, discontinuation of a concomitantly used cytochrome P450 3A4 inducer may result in an increase in fentanyl plasma concentration. Evaluate patients at frequent intervals receiving fentanyl and any CYP3A4 inhibitor or inducer.

Neonatal opioid withdrawal syndrome:

Advise pregnant women using opioids for an extended period of time of the risk of neonatal opioid withdrawal syndrome (NOWS), which may be life-threatening if not recognized and treated. Ensure the management by neonatology experts will be available at delivery.

Risks from concomitant use with benzodiazepines or other CNS depressants:

Concomitant use of opioids with benzodiazepines or other CNS depressants, including alcohol, may result in profound sedation, respiratory depression, coma, and death. Reserve concomitant prescribing of fentanyl and benzodiazepine or other CNS depressants for use in patients for whom alternative treatment options are inadequate.

Risk of medication errors (buccal, intranasal, lozenge, sublingual):

Substantial differences exist in the pharmacokinetic profile of fentanyl buccal, intranasal, lozenge, and sublingual compared with other fentanyl products that result in clinically important differences in the extent of absorption of fentanyl that could result in fatal overdose. When prescribing, do not convert patients on a mcg-per-mcg basis from any other fentanyl products to fentanyl buccal, intranasal, lozenge, or sublingual. When dispensing, do not substitute a fentanyl buccal, intranasal, lozenge, or sublingual prescription for other fentanyl products.

REMS program:

Buccal, intranasal, lozenge, sublingual: Because of the risk for accidental exposure, misuse, abuse, addiction, and overdose, fentanyl buccal, intranasal, lozenge, and sublingual are available only through a restricted program required by the FDA, called a Risk Evaluation and Mitigation Strategy (REMS). Under the Transmucosal Immediate Release Fentanyl (TIRF) REMS, pharmacies, outpatients, and health care professionals who prescribe to outpatients must enroll in the program. Inpatient pharmacies must develop policies and procedures to verify opioid tolerance in inpatients who require fentanyl buccal, intranasal, lozenge, and sublingual while hospitalized. Further information is available at <http://www.TIRFREMSaccess.com> or by calling 1-866-822-1483.

Transdermal patch: Health care providers are strongly encouraged to complete a REMS-compliant education program and to counsel patients and caregivers on serious risks, safe use, and the importance of reading the Medication Guide with each prescription.

Exposure to heat (transdermal patch only):

Exposure of the fentanyl application site and surrounding area to direct external heat sources, such as heating pads or electric blankets, heat or tanning lamps, sunbathing, hot baths, saunas, hot tubs, and heated water beds may increase fentanyl absorption and has resulted in fatal overdose of fentanyl.

Warn patients to avoid exposing the application site and surrounding area to direct external heat sources.

Brand Names: US

- Actiq [DSC];
- Fentora [DSC];
- Lazanda [DSC];
- Subsys [DSC]

Brand Names: Canada

- Duragesic 50 [DSC];
- FentaNYL Citrate SDZ;
- Fentora;
- MYLAN-FentaNYL Matrix [DSC];
- PMS-FentaNYL MTX [DSC];
- SANDOZ FentaNYL;
- TEVA FentaNYL;
- TEVA-FentaNYL

Pharmacologic Category

- Analgesic, Opioid;
- Anilidopiperidine Opioid;
- General Anesthetic

Dosing: Adult

Dosage guidance:

Safety: Consider prescribing an opioid overdose reversal agent (eg, naloxone, nalmefene) for patients with factors associated with an increased risk for overdose, such as history of overdose or substance use disorder, patients with sleep-disordered breathing, higher opioid dosages (≥ 50 morphine milligram equivalents [MME]/day orally), and/or concomitant benzodiazepine use (Ref).

Dosing: Dosing provided is based on typical doses; some patients may require higher or lower doses. Individualize dosing based on patient-specific factors (eg, severity of pain, comorbidities, degree of opioid experience/tolerance) and titrate to patient-specific treatment goals (eg, improvement in function and quality of life, decrease in pain using validated pain rating scale). Use the lowest effective dose for the shortest period of time.

Clinical considerations: Opioids may be part of a comprehensive, multimodal, patient-specific treatment plan for managing severe pain. Maximize nonopioid analgesia (when appropriate) prior to initiation of opioid analgesia (Ref).

Acute pain, patient-controlled analgesia

Acute pain, patient-controlled analgesia (alternative agent): Note: Generally, fentanyl is the preferred opioid for patients with severe kidney or hepatic dysfunction and/or for patients who are unable to tolerate morphine or hydromorphone (Ref). **For use in ICU, postoperative, or other closely monitored settings.**

IV:

Fentanyl: Example PCA Initial Dose Ranges^a

^a PCA = patient-controlled analgesia. For use to maintain pain control after initial pain control achieved. If analgesia is inadequate, may adjust dosing and provide rescue bolus doses (Buys 2026).

^b If patient is chronically taking high doses of opioids (eg, >50 morphine milligram equivalents per day) or medication for opioid use disorder (ie, methadone or buprenorphine), may initiate with demand doses at the high end of the range and a lockout interval at the low end of the range, while continuing their baseline opioid; monitor such patients closely for pain control and opioid-related side effects and adjust the PCA settings accordingly (Buys 2026).

^c Due to the risk of respiratory depression, use of a basal dose should be limited to carefully selected patients who are opioid tolerant and/or receiving care in a critical care unit (Buys 2026; ISMP 2009).

Usual concentration

10 mcg/mL

Demand dose^b

Usual range: 5 to 20 mcg

Basal dose

In general, a continuous (basal) infusion is **not** recommended in an opioid-naïve patient^c

Lockout interval^b

4 to 8 minutes

Acute postoperative pain

Acute postoperative pain:

Postoperative recovery/Postanesthesia care unit (ie, immediate postoperative period):

IV: 25 to 50 mcg every 5 minutes (moderate pain) or 50 to 100 mcg every 2 to 5 minutes (severe pain) until pain is relieved or unwanted side effects appear; after initial pain control, readdress postoperative analgesic regimen to optimize comfort (Ref).

IM: 50 to 100 mcg every 1 to 2 hours as needed. **Note:** IM route should only be used if IV administration is not available (eg, loss of IV access).

Acute nonoperative severe pain

Acute nonoperative severe pain:

Note: For use in ICU or other closely monitored settings. Do not prescribe fentanyl for use on an outpatient basis for acute noncancer-related pain; consider the use of other oral opioids (Ref). Dose modifications may be warranted based on patient-specific factors (eg, age, expected degree of pain, weight) (Ref).

Intermittent dosing:

IV, IM: 25 to 50 mcg **or** 0.35 to 0.5 mcg/kg every 30 to 60 minutes as needed (Ref). **Note:** More frequent administration may be necessary when used by the IV route due to short duration of activity. IM route should only be used if IV administration is not available (eg, loss of IV access).

SUBQ (off-label route) (alternative agent): 25 to 50 mcg every 1 to 2 hours as needed (Ref).

Note: Specific data on SUBQ intermittent dosing is not available; adjust dose according to patient response under close monitoring. If usual dose and frequency is insufficient, reassess patient and reconsider pain management strategies. Refer to institutional protocols.

Chronic pain, including chronic cancer pain

Chronic pain, including chronic cancer pain:

Note: Opioids, including fentanyl, are **not** the preferred therapy for chronic noncancer pain due to insufficient evidence of benefit and risk of serious harm; nonpharmacologic treatment and nonopioid analgesics are preferred, with the exception of chronic pain from sickle cell disease and end-of-life care. Opioids, including fentanyl, should **only** be considered in patients with chronic, noncancer pain who are expected to experience clinically meaningful improvement in pain and function that outweighs patient safety risks (Ref).

SUBQ continuous infusion: Note: For progressive illnesses (eg, cancer), a continuous SUBQ infusion, with or without a patient-controlled analgesia option, can be used as pain requirements increase. In general, SUBQ continuous infusion dose is equivalent to IV continuous infusion dose (Ref). Individualize dose based on previous opioid intake and appropriate opioid analgesic equivalents; titrate

further, if needed, based on level of pain. Reported dosing varies greatly and is based on patient needs; refer to institutional protocols (Ref).

Transdermal patch: Discontinue or taper all other around-the-clock or extended-release opioids when initiating therapy with fentanyl transdermal patch.

Initial: To convert patients from oral or parenteral opioids to fentanyl transdermal patch, a 24-hour analgesic requirement should be calculated (based on prior opioid use). Using the table below, the appropriate initial dose can be determined. The initial fentanyl dosage may be approximated from the 24-hour morphine dosage equivalent and titrated to minimize adverse effects and provide analgesia. While there are useful tables of opioid equivalents available, substantial interpatient variability exists in relative potency of different opioids and products. Therefore, it is safer to underestimate the daily fentanyl requirement and provide breakthrough pain relief with rescue medication (eg, immediate-release opioid) than to overestimate requirements, which could result in adverse reactions. With the initial application, the absorption of transdermal fentanyl requires several hours to reach plateau; therefore, transdermal fentanyl is inappropriate for management of acute pain. Change patch every 72 hours. The majority of patients may be controlled on every-72-hour administration; however, some patients may require every-48-hour administration because of more breakthrough pain in the last 24 hours of each cycle.

Conversion from continuous IV infusion of fentanyl: In patients who have adequate pain relief with IV fentanyl infusion, may convert to transdermal dosing at a rate equivalent to the IV rate using a 2-step taper of the infusion to be completed over 12 hours after the patch is applied. Six hours after the application of the first patch, decrease the infusion to 50% of the original rate; discontinue infusion 12 hours after patch application (Ref).

Titration: Do not titrate more frequently than every 3 days after the initial application or every 6 days thereafter. Short-acting opioids may be required until analgesic efficacy is established and/or as supplements for breakthrough pain. The number and quantity of supplemental doses should be closely monitored. When increasing the dose, base the new dose on the daily requirement of supplemental opioids required by the patient during the second or third day of initial application. For example, if 24-hour oral morphine requirement for breakthrough pain is 50 mg, then may increase transdermal fentanyl dose by 25 mcg/hour (Ref).

Dose conversion guidelines for transdermal fentanyl (see table below):

Note: Using the manufacturer's recommended dose conversion guidelines, based upon the daily oral morphine dose, may underestimate the transdermal fentanyl strength required and result in the need for supplemental immediate-release opioid therapy for breakthrough pain or in the patient experiencing withdrawal syndrome (Ref). The manufacturers recommend a ratio of approximately 45 mg/24 hours of oral morphine to a 12 mcg/hour fentanyl dosage (US labeling) or the ratio of 45 to 59 mg/24 hours of oral morphine to a 12 mcg/hour fentanyl dosage (Canadian labeling). Below is a less conservative dosing conversion strategy based on a 2:1 ratio of oral morphine to transdermal fentanyl (Ref). For the more conservative dose conversion strategy, see the manufacturer's labeling. The table is only to be used for the conversion from current opioid therapy to transdermal fentanyl. **Do not use this table to convert from transdermal fentanyl to another opioid (doing so may lead to fatal overdose due to overestimation of the new opioid).** This is **not** a table of equianalgesic doses.

Step 1: Determine the patient's 24-hour oral morphine requirement. If patient was not receiving oral morphine, must convert the 24-hour requirement to the oral morphine equivalent using a conversion chart or tool.

Step 2: Once the 24-hour oral morphine requirement is determined, use the Dose Conversion Guidelines to determine the appropriate fentanyl transdermal dose (mcg/hour).

Dose Conversion Guidelines: Recommended Initial Fentanyl Transdermal Dose Based Upon Daily Oral Morphine Dose^{a,b,c}

Oral 24-Hour Morphine Dose (mg/day)

Fentanyl Transdermal Dose (mcg/hour)

^a Portenoy 2019a.

^b The table should **not** be used to convert from transdermal fentanyl to other opioid analgesics. Rather, following removal of the patch, titrate the dose of the new opioid until adequate analgesia is achieved.

^c Suggested doses for conversion to transdermal fentanyl from other opioids are less conservative than recommendations in the US product labeling. The recommendations in this table are based on guidance available at experienced centers.

25

12

50

25

100

50

150

75

200

100

250

125

300

150

350

175

400

200

450

225

500

250

550

275

600

300

Discontinuation or tapering of therapy :

When reducing the dose, discontinuing, or tapering long-term opioid therapy, the dose should be gradually tapered. An optimal tapering schedule has not been established. Individualize tapering based on discussions with patient to minimize withdrawal, while considering patient-specific goals and concerns and the opioid's pharmacokinetics. Proposed initial schedules range from slow (eg, 10% reduction per week or 10% reduction per month depending on duration of long-term therapy) to rapid (eg, 25% to 50% reduction every few days) (Ref). When discontinuing transdermal fentanyl and not converting to another opioid, particularly in patients who are physically opioid dependent,

use a gradual downward titration (eg, decrease the dose by 25% every 2 to 4 weeks) (Ref). Upon system removal of transdermal fentanyl, ≥ 17 hours are required for a 50% decrease in fentanyl levels. Slower tapers may be appropriate after long-term use (eg, >1 year), whereas more rapid tapers may be appropriate in patients experiencing severe adverse effects. During tapering, patients may be at an increased risk of overdose if they return to their original (or higher) opioid dose or use illicit opioids, due to rapid loss of tolerance; consider prescribing an opioid overdose reversal agent (eg, naloxone, nalmeferene). Monitor carefully for signs/symptoms of withdrawal. If the patient displays withdrawal symptoms, consider slowing the taper schedule; alterations may include increasing the interval between dose reductions, decreasing amount of daily dose reduction, pausing the taper and restarting when the patient is ready, and/or coadministration of an alpha-2 agonist (eg, clonidine) to blunt autonomic withdrawal symptoms and other adjunctive agents to treat GI symptoms and muscle spasms, as needed. Continue to offer nonopioid analgesics as needed for pain management during the taper (Ref). In patients who continue to take long-term opioid therapy but no longer require fentanyl for breakthrough pain, fentanyl can usually be discontinued without a taper.

Chronic cancer pain, management of breakthrough pain episodes

Chronic cancer pain, management of breakthrough pain episodes:

Transmucosal:

Note: For patients who are tolerant to and currently receiving opioid therapy for persistent cancer pain; dosing should be individually titrated to provide adequate analgesia with minimal side effects. Dose titration should be done if patient requires more than 1 dose per breakthrough pain episode for several consecutive episodes. Patients experiencing >4 breakthrough pain episodes per day should have the dose of their long-term opioid reevaluated. **Patients must remain on around-the-clock opioids during use** (Ref).

Lozenge (Actiq): Note: Fentanyl lozenge (brand and generic) has been discontinued in the United States >1 year. Do **not** convert patients from any other fentanyl product to Actiq on a mcg-per-mcg basis. Patients previously using another fentanyl product should be initiated at a dose of 200 mcg; individually titrate to provide adequate analgesia while minimizing adverse effects.

Initial dose: 200 mcg (consumed over 15 minutes) for all patients; if after 30 minutes from the start of the lozenge (ie, 15 minutes following the completion of the lozenge) pain is unrelieved, a second 200 mcg dose may be given over 15 minutes. A maximum of 1 additional dose can be given per pain episode; **must wait at least 4 hours before treating another episode**. To limit the number of units in the home during titration, only prescribe an initial titration supply of six 200 mcg lozenges.

Dose titration: From the initial dose, closely monitor patients and modify the dose until patient reaches a dose providing adequate analgesia using a single dosage unit per breakthrough cancer pain episode. If signs/symptoms of excessive opioid effects (eg, respiratory depression) occur, immediately remove the dosage unit from the patient's mouth, dispose of properly, and reduce subsequent doses. If adequate relief is not achieved 15 minutes after completion of the first dose (ie, 30 minutes after the start of the lozenge), only 1 additional lozenge of the same strength may be given for that episode; **must wait at least 4 hours before treating another episode**.

Maintenance dose: Once titrated to an effective dose, patients should generally use a single dosage unit per breakthrough pain episode. During any pain episode, if adequate relief is not achieved 15 minutes after completion of the first dose (ie, 30 minutes after the start of the lozenge), only 1 additional lozenge of the same strength may be given over 15 minutes for that episode; **must wait at least 4 hours before treating another episode**. Consumption should be limited to ≤ 4 units per day (once an effective breakthrough dose is found). If adequate analgesia is **not** provided after treating several episodes of breakthrough pain using the same dose, increase dose to next highest lozenge strength (initially dispense no more than 6 units of the new strength). Reevaluate the around-the-clock opioid therapy in patients experiencing >4 breakthrough pain episodes per day. If signs/symptoms of excessive opioid effects (eg, respiratory depression) occur, immediately remove the dosage unit from the patient's mouth, dispose of properly, and reduce subsequent doses.

Buccal tablet (Fentora): Note: Fentanyl buccal tablets (brand and generic) have been discontinued in the United States for >1 year. Do **not** convert patients from any other fentanyl product

to Fentora on a mcg-per-mcg basis. Patients previously using another fentanyl product should be initiated at a dose of 100 mcg (except Actiq); individually titrate to provide adequate analgesia while minimizing adverse effects. For patients previously using the transmucosal lozenge (Actiq), the initial dose should be selected using the conversions listed; see "Conversion From Lozenge (Actiq) to Buccal Tablet (Fentora)".

Initial dose: 100 mcg for all patients unless patient is already using Actiq; see "Conversion From Lozenge (Actiq) to Buccal Tablet (Fentora)". If after 30 minutes pain is unrelieved, a second 100 mcg dose may be administered (US labeling) or an alternative analgesic rescue medication (other than Fentora) may be given (Canadian labeling). A maximum of 2 doses (or 1 dose [Canadian labeling]) can be given per breakthrough pain episode. **Must wait at least 4 hours before treating another episode with buccal tablet.**

Dose titration: If titration required, the 100 mcg dose may be increased to 200 mcg using two 100 mcg tablets (one on each side of mouth) with the next breakthrough pain episode. If 200 mcg dose is not successful, patient can use four 100 mcg tablets (two on each side of mouth) with the next breakthrough pain episode. If titration requires >400 mcg per dose, titrate using 200 mcg tablets; do not use more than 4 tablets simultaneously (maximum single dose: 800 mcg). During any breakthrough pain episode, if adequate relief is not achieved 30 minutes after buccal tablet application, a second dose of same strength for that breakthrough pain episode may be used (US labeling) or an alternative analgesic rescue medication (other than Fentora) may be given (Canadian labeling). **Must wait at least 4 hours before treating another episode with buccal tablet.**

Maintenance dose: Following titration, the effective maintenance dose using 1 tablet of the appropriate strength should be administered once per episode; if after 30 minutes pain is unrelieved, may administer a second dose of the same strength (US labeling) or an alternative analgesic rescue medication (other than Fentora) may be given (Canadian labeling). **Must wait at least 4 hours before treating another episode with buccal tablet.** Limit to 4 applications per day. Reevaluate the around-the-clock opioid therapy in patients experiencing >4 breakthrough pain episodes per day. Once an effective maintenance dose has been established, the buccal tablet may be administered sublingually (alternative route). To prevent confusion, patient should only have one strength available at a time. Once maintenance dose is determined, all other unused tablets should be disposed of and that strength (using a single tablet) should be used. Using more than 4 buccal tablets at a time has not been studied.

Conversion from lozenge (Actiq) to buccal tablet (Fentora):

Lozenge dose 200 or 400 mcg: Initial buccal tablet dose is 100 mcg; may titrate using multiples of 100 mcg.

Lozenge dose 600 or 800 mcg: Initial buccal tablet dose is 200 mcg; may titrate using multiples of 200 mcg.

Lozenge dose 1,200 or 1,600 mcg: Initial buccal tablet dose is 400 mcg (using two 200 mcg tablets); may titrate using multiples of 200 mcg.

Intranasal (Lazanda): Note: Lazanda has been discontinued in the United States for >1 year. Do **not** convert patients from any other fentanyl product to Lazanda on a mcg-per-mcg basis. Patients previously using another fentanyl product should be initiated at a dose of 100 mcg (except Actiq); individually titrate to provide adequate analgesia while minimizing adverse effects.

Initial dose: 100 mcg (one 100 mcg spray in one nostril) for all patients; if after 30 minutes pain is unrelieved, an alternative rescue medication may be used. **Must wait at least 2 hours before treating another episode with fentanyl intranasal.** However, for the next pain episode, increase to a higher dose using the recommended dose titration steps.

Dose titration: If titration required, increase to a higher dose for the next pain episode using the following titration steps (**Note: Must wait at least 2 hours before treating another episode with fentanyl intranasal**): If no relief with 100 mcg dose, increase to 200 mcg dose per episode (one 100 mcg spray in each nostril); if no relief with 200 mcg dose, increase to 300 mcg dose per episode (alternating one 100 mcg spray in right nostril, one 100 mcg spray in left nostril, and one

100 mcg spray in the right nostril); if no relief with 300 mcg dose, increase to 400 mcg dose per episode (one 400 mcg spray in one nostril or alternating two 100 mcg sprays in each nostril); if no relief with 400 mcg dose, increase to 600 mcg dose per episode (one 300 mcg spray in each nostril); if no relief with 600 mcg dose, increase to 800 mcg dose per episode (one 400 mcg spray in each nostril). **Note:** Single doses >800 mcg have not been evaluated. Avoid use of a combination of dose strengths to treat an episode, as this may cause confusion and dosing errors.

Maintenance dose: Once maintenance dose for breakthrough pain episode has been determined, use that dose for subsequent episodes. For pain that is not relieved 30 minutes after Lazanda administration or if a separate breakthrough pain episode occurs within the 2-hour window before the next Lazanda dose is permitted, a rescue medication may be used. Limit Lazanda use to ≤ 4 doses per day. If patient is experiencing >4 breakthrough pain episodes per day, consider increasing the around-the-clock, long-acting opioid therapy; if long-acting opioid therapy dose is altered, reevaluate and retitrate Lazanda dose as needed. If response to maintenance dose changes (increase in adverse reactions or alterations in pain relief), dose readjustment may be necessary.

Sublingual:

Sublingual spray (Subsys): Note: Subsys has been discontinued in the United States for >1 year. Do **not** convert patients from any other fentanyl product to Subsys on a mcg-per-mcg basis. Patients previously using another fentanyl product should be initiated at a dose of 100 mcg (except Actiq); individually titrate to provide adequate analgesia while minimizing adverse effects. For patients previously using the transmucosal lozenge (Actiq), the initial dose should be selected using the conversions listed; see "Conversion From Lozenge (Actiq) to Sublingual Spray (Subsys)".

Initial dose: 100 mcg for all patients unless patient is already using Actiq; see "Conversion From Lozenge (Actiq) to Sublingual Spray (Subsys)". If after 30 minutes pain is unrelieved, 1 additional 100 mcg dose may be given. A maximum of 2 doses can be given per breakthrough pain episode.

Must wait at least 4 hours before treating another episode with sublingual spray.

Dose titration: If titration required, titrate to a dose that provides adequate analgesia (with tolerable side effects) using the following titration steps: If no relief with 100 mcg dose, increase to 200 mcg dose (using one 200 mcg unit); if no relief with 200 mcg dose, increase to 400 mcg dose (using one 400 mcg unit); if no relief with 400 mcg dose, increase to 600 mcg dose (using one 600 mcg unit); if no relief with 600 mcg dose, increase to 800 mcg dose (using one 800 mcg unit); if no relief with 800 mcg dose, increase to 1,200 mcg dose (using two 600 mcg units); if no relief with 1,200 mcg dose, increase to 1,600 mcg dose (using two 800 mcg units). During dose titration, if breakthrough pain is unrelieved 30 minutes after Subsys administration, 1 additional dose using the same strength may be administered (maximum: 2 doses per breakthrough pain episode); **patient must wait 4 hours before treating another breakthrough pain episode with sublingual spray.**

Maintenance dose: Once maintenance dose for breakthrough pain episode has been determined, use that dose for subsequent episodes. If occasional episodes of unrelieved breakthrough pain occur 30 minutes after Subsys administration, 1 additional dose using the same strength may be administered (maximum: 2 doses per breakthrough pain episode); **patient must wait 4 hours before treating another breakthrough pain episode with sublingual spray.** Once maintenance dose is determined, limit Subsys use to ≤ 4 doses per day. If response to maintenance dose changes (increase in adverse reactions or alterations in pain relief), dose readjustment may be necessary. If patient is experiencing >4 breakthrough pain episodes per day, reevaluate the around-the-clock, long-acting opioid therapy.

Conversion from lozenge (Actiq) to sublingual spray (Subsys):

Lozenge dose 200 or 400 mcg: Initial sublingual spray dose is 100 mcg; may titrate using multiples of 100 mcg.

Lozenge dose 600 or 800 mcg: Initial sublingual spray dose is 200 mcg; may titrate using multiples of 200 mcg.

Lozenge dose 1,200 or 1,600 mcg: Initial sublingual spray dose is 400 mcg; may titrate using multiples of 400 mcg.

Critically ill patients in the ICU, analgesia and sedation

Critically ill patients in the ICU, analgesia and sedation (off-label use):

Note: Multimodal approaches (eg, a combination of analgesics and techniques) should typically be employed for pain control in this setting. Pain should be monitored using validated scales (eg, behavioral pain scale, critical-care pain observation tool) in ICU patients who are unable to self-report (Ref).

Intermittent dosing:

Loading dose: **IV:** 25 to 100 mcg **or** 1 to 2 mcg/kg; may repeat dose if severe pain persists and adverse effects are minimal at the time of expected peak effect (eg, ~5 minutes after administration) (Ref). Follow with intermittent maintenance dose or a continuous infusion.

Maintenance dose: **IV:** 25 to 50 mcg **or** 0.35 to 0.5 mcg/kg every 30 to 60 minutes as needed (Ref).

Continuous infusion:

IV: After initial loading dose, begin continuous infusion at an initial rate of 25 to 50 mcg/hour; titrate every 30 to 60 minutes to clinical effect (ie, pain control and/or sedation). May administer an additional small bolus dose (eg, 25 mcg) prior to increasing the infusion rate (Ref). Usual dosing range: 50 to 300 mcg/hour; weight-based dosing range: 0.7 to 10 mcg/kg/hour (Ref). **Note:** Fentanyl can accumulate in lipid stores when used for extended periods of time and may result in prolonged sedation and reduced ability to liberate from mechanical ventilator (Ref).

Procedural sedation and analgesia

Procedural sedation and analgesia:

Outside the operating room (alternative agent) (off-label use) :

IV: 0.5 to 1 mcg/kg every 2 minutes until desired level of sedation and analgesia achieved (Ref); generally, the maximum total dose is 250 mcg. If administered with other sedatives (eg, etomidate, propofol, midazolam), do not exceed single doses of 0.5 mcg/kg (Ref).

Intranasal (using parenteral solution [50 mcg/mL]) (off-label route): 1 to 2 mcg/kg as a single dose; administer half the dose in each nostril; maximum total dose: 100 mcg (50 mcg per nostril; based on volume) (Ref). **Note:** Recommend using a nasal atomizer for delivery (Ref).

Analgesia during monitored anesthesia care or regional anesthesia: **IV:** Usual initial dose range: 0.5 to 2 mcg/kg, administered in incremental boluses of 25 to 50 mcg, titrated to effect. When used in combination with a sedative (eg, midazolam), consider dosage reduction (Ref). **Note:** Since an IV should be in place with anesthesia, the IM route is rarely used but still maintained as an option in the manufacturer's labeling. If IM route is used, the dose is equivalent to the recommended IV dose.

Rapid sequence intubation, pretreatment

Rapid sequence intubation, pretreatment (off-label use): **IV:** Usual dose: 50 to 200 mcg (or 1 to 3 mcg/kg) over 30 to 60 seconds administered ~3 minutes prior to induction. In patients with tenuous hemodynamic status, avoid use or use lower doses (eg, 1 mcg/kg [or 50 mcg]); in patients with elevated intracranial pressure, use higher doses (eg, 3 mcg/kg [or 200 mcg]). Avoid use in patients with shock or significant hemodynamic instability (Ref).

General anesthesia

General anesthesia:

Preinduction: **IV:** 25 mcg; may repeat in increments of 25 mcg (typical total dose is ≤100 mcg) to provide pain relief or if patient requires a regional anesthesia procedure prior to surgery (Ref).

Induction: **IV:** 25 to 100 mcg (or 0.5 to 1 mcg/kg) (Ref).

Maintenance: **Note:** When used as part of multimodal analgesia within enhanced recovery after surgery protocols, some experts recommend administering no more than 1 mcg/kg/hour (Ref).

Supplemental analgesia during maintenance of general anesthesia with inhaled agents:
Intermittent: **IV:** 25 to 50 mcg bolus as needed (Ref).

Supplement to total IV anesthesia (TIVA) when controlled postoperative ventilation is planned: Continuous infusion: **IV:** 1 to 2 mcg/kg/hour (Ref).

Neuraxial analgesia

Neuraxial analgesia:

Epidural: Note: Reserve use for patients with severe acute pain (eg, after major abdominal surgery, cancer pain, during labor and delivery). Must be administered by health care providers skilled in the care of patients receiving intraspinal opioids (Ref). Use a **preservative-free** formulation intended for neuraxial use. Example regimens and usual doses are provided; specific regimens and doses vary based on type of surgery, comorbidities, and institutional policies and practices.

Single dose: 25 to 100 mcg; may provide adequate relief for up to 8 hours. May repeat with additional 100 mcg boluses on demand (Ref) or alternatively may administer by a continuous infusion (Ref).

Continuous infusion: 25 to 100 mcg/hour (fentanyl alone). When combined with a local anesthetic (eg, bupivacaine), fentanyl requirement is less (Ref).

Intrathecal: Note: Reserve use for patients with severe acute pain (eg, after major abdominal surgery, cancer pain, during labor and delivery). Must be administered by health care providers skilled in the care of patients receiving intraspinal opioids (Ref). Use a **preservative-free** formulation intended for neuraxial use.

Single dose: 15 to 25 mcg; may provide adequate relief for up to 6 hours. When combined with a local anesthetic (eg, bupivacaine), fentanyl dose requirement is less (eg, 10 to 15 mcg instead of 15 to 25 mcg) (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Note: Although limited pharmacokinetic data exist in patients with renal insufficiency, <7% to 10% of fentanyl is excreted as unchanged drug and its metabolites are inactive. Fentanyl may be used and is generally considered safe for use in patients with kidney impairment with low initial doses and careful monitoring for accumulation and adverse effects (Ref).

Altered kidney function:

Injection: CrCl <50 mL/minute:

No dosage adjustment necessary when single or infrequent bolus doses are used for short interventional procedures (Ref). For more frequent dosing, use small, incremental doses to titrate based on analgesic response and adverse effects (Ref). May need to decrease dose to avoid accumulation (Ref), especially with continuous infusions (Ref).

Transdermal (patch):

Note: Although the manufacturer recommends to avoid use of transdermal fentanyl in severe impairment, fentanyl can be used safely in these patients with caution and close monitoring with initial and long-term use (Ref). The following dose adjustments have been proposed when initiating transdermal fentanyl:

CrCl >50 mL/minute: No dosage adjustment necessary.

CrCl 10 to 50 mL/minute: Initial: 75% of normal dose (Ref).

CrCl <10 mL/minute: Initial: 50% of normal dose (Ref).

Transmucosal (buccal tablet, sublingual spray, lozenge) and intranasal: There are no dosage adjustments provided in the manufacturer's labeling; use with caution. Although fentanyl pharmacokinetics may be altered in kidney disease, transmucosal fentanyl can be used successfully in the management of breakthrough cancer pain. Doses should be titrated to reach clinical effect with careful monitoring of patients with severe kidney disease.

Dialysis, intermittent (thrice weekly): Not dialyzable (Ref):

Transdermal patch: Initial: 50% of normal dose (Ref).

All other formulations/routes: Use lowest effective dose with gradual titration and close monitoring of response and adverse effects (Ref).

Peritoneal dialysis:

Transdermal patch: Initial: 50% of normal dose (Ref).

All other formulations/routes: Use lowest effective dose with gradual titration and close monitoring of response and adverse effects (Ref).

CRRT:

All formulations/routes: Use lowest effective dose with gradual titration and close monitoring of response and adverse effects (Ref).

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Dosage guidance:

Clinical considerations: Fentanyl has a high first pass liver extraction ratio, and therefore, clearance will be reduced in the setting of impaired hepatic blood flow as opposed to reduced hepatic 3A4 enzyme activity or protein binding (Ref).

Liver impairment prior to treatment initiation:

Initial or dose adjustment in patients with preexisting liver cirrhosis:

Injection:

Child-Turcotte-Pugh class A and B: No dosage adjustment necessary (Ref).

Child-Turcotte-Pugh class C:

Note: The following recommendations are considered applicable for all indications where intermittent dosing of injectable fentanyl may be utilized (eg, procedural analgesia and/or sedation). In general, avoid IM, continuous IV infusions, and SUBQ administration. Use the IM route only if IV administration is unavailable (eg, loss of IV access) (Ref).

Initial or loading dose: Administer 50% of the usual dose (Ref).

Repeat or maintenance dose : Administer 50% of the usual dose. May also extend the dosing interval by 1.5 to 2 times the usual dosing interval (eg, if the usual dosing interval is every 1 to 2 hours then extend the interval to ~2 to 4 hours); individualize dose based on the benefits and the risks; use of intermittent (ie, as needed) versus scheduled dosing is preferred (Ref).

Transdermal (patch): Note: Avoid use in patients with cirrhosis and concurrent cachexia, sarcopenia, or fever (Ref).

Child-Turcotte-Pugh class A and B: Initial: Administer 50% of the usual dose (Ref).

Child-Turcotte-Pugh class C: Maximum dose: A single transdermal patch that delivers 12.5 mcg/hour; do not apply more often than every 72 hours (Ref).

Transmucosal (buccal tablet, sublingual spray, lozenge) and intranasal:

Child-Turcotte-Pugh class A and B: Initiate with lowest dose available based on dosage form; may utilize the usual adult dosing recommendations for dose titration and maintenance dose along with frequent monitoring for adverse effects (Ref).

Child-Turcotte-Pugh class C: Initiate with lowest dose available based on dosage form and extend dosing interval by at least 1.5 to 2 times the usual dosing interval (eg, if the usual dosing interval is ≥ 4 hours then increase the interval to ≥ 6 hours) weighing the benefits and the risks of fentanyl accumulation; use of intermittent (ie, as needed) versus scheduled dosing is preferred (Ref).

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1, 2, or 3 obesity (BMI ≥ 30 kg/m²):

IV: Use a standard, non-weight-based dose (based on indication) in the majority of patients (eg, pain, ICU analgesia/sedation), then titrate to clinical effect (Ref). If weight-based doses are utilized, use **adjusted body weight** for initial weight-based dose calculations (based on indication), then titrate to clinical effect (Ref). Clinicians should **not** change dosing weight from one weight metric to another during therapy (ie, adjusted body weight to/from actual body weight) (Ref). Refer to adult dosing for indication-specific doses.

Non-IV route (eg, transdermal, IM, intranasal): Use a standard non-weight-based dose (based on indication), then titrate to clinical effect (Ref). Refer to adult dosing for indication-specific doses.

Rationale for recommendations: Studies evaluating other opioids in patients with obesity have shown large variations in opioid requirements with no relationship to actual body weight (Ref). Pharmacokinetic studies evaluating fentanyl, a highly lipophilic compound, have shown that if actual body weight is used for calculating dosing in patients with obesity, this may lead to excessive fentanyl concentrations and accumulation with continued use due to reduced clearance as actual body weight increases (Ref). Therefore, use of an alternative size descriptor is recommended for dosing fentanyl in patients with obesity.

Guidelines have suggested using lean body weight (LBW) for initial weight-based dosing of opioids, titrated to effect (Ref); however, there are concerns with calculation errors when using the complex LBW formula, especially in critical situations (Ref). There are also concerns with underdosing in clinical situations where there is a need to optimize drug concentrations (eg, procedural sedation). Use of an alternative size descriptor (ie, adjusted body weight) is recommended for weight-based dosing in patients with class 1, 2, or 3 obesity (BMI ≥ 30 kg/m²) (Ref).

Dosing: Older Adult

Note: Minimize opioid use in older adults unless for the management of severe acute pain. Opioids are associated with an increased risk of falls and inducing or worsening delirium in older adults (Ref). Elderly patients have been found to be twice as sensitive as younger patients to the effects of fentanyl. A wide range of doses may be used. When choosing a dose, take into consideration the following patient factors: age, weight, physical status, underlying disease states, other drugs used, type of anesthesia used, and the surgical procedure to be performed.

Transmucosal lozenge (eg, Actiq): In clinical trials, patients who were >65 years of age were titrated to a mean dose that was 200 mcg less than that of younger patients.

Dosing: Pediatric

(For additional information see "Fentanyl: Pediatric drug information")

Dosage guidance:

Dosing: Titrate dose to appropriate effects; wide range of doses exist, dependent upon desired degree of analgesia/anesthesia, clinical environment, patient's status, and presence of opioid tolerance.

Route of administration is variable (eg, IV, intranasal, epidural); use caution.

Acute pain, severe

Acute pain, severe (opioid-naïve):

Intermittent IV doses:

Infants, Children, and Adolescents: Limited data available in <2 years of age:

Patient weight <50 kg: IV: Initial: 0.5 to 1 mcg/kg/dose; may repeat every 1 to 2 hours; more frequent dosing may be allowed (eg, every 30 minutes) if necessary; usual maximum dose: 50 mcg/dose; a higher maximum dose of 100 mcg/dose may be considered in critically ill patients in the ICU (Ref).

Patient weight ≥50 kg: Initial: IV: 25 to 50 mcg; repeat every 1 to 2 hours (more frequent dosing may be needed [eg, every 30 minutes]) (Ref).

Continuous IV infusion:

Infants, Children, and Adolescents:

Patient weight <50 kg: Limited data available: Continuous IV infusion: Usual range: 0.5 to 2.5 mcg/kg/**hour**; initiate at the lower end and titrate to effect; an initial rate of 1 mcg/kg/**hour** has been suggested for critically ill patients in the ICU (Ref).

Patient weight ≥50 kg: Limited data available: Continuous IV infusion: Usual range: 25 to 100 mcg/**hour**; initiate at the lower end of dosage range and titrate to effect; an initial dose of 50 mcg/**hour** has been suggested for critically ill patients in the ICU (Ref).

Intranasal (using parenteral preparation): Limited data available: Children ≥10 kg and Adolescents: Intranasal: Usual dose: 1.5 to 2 mcg/kg once; maximum dose: 100 mcg/dose (Ref); some studies that used an initial dose of 1.5 mcg/kg allowed for second dose of 0.5 mcg/kg or 0.75 mcg/kg, at the discretion of the physician, 10 or 20 minutes, respectively, after the first dose (Ref).

Anesthesia

Anesthesia:

Note: Dosing presented reflects general ranges; specific dose dependent on patient characteristics, procedure, and/or other anesthetic agents employed.

Induction and maintenance: Children 2 to 12 years: IV: 2 to 10 mcg/kg/dose; followed by a continuous infusion in some cases at 2 to 5 mcg/kg/**hour** (Ref).

Anesthesia, general; adjunct: **Note:** Specific dose dependent on type of procedure/surgery and other anesthetics in use. Dosing provided is a general range and dose should be individualized. Specific maintenance and total dosages should be based on intraoperative clinical response managed by clinicians trained in anesthesia: Adolescents: IV: 0.5 to 50 mcg/kg/dose.

Anesthesia; regional

Anesthesia, regional: Limited data available:

Note: Use only preservative-free preparations. Dosing presented reflects general ranges; specific dose dependent on patient characteristics, procedure, and/or other anesthetic agents employed.

Infants, Children, and Adolescents: Epidural catheter: Continuous epidural infusion: 0.3 to 1 mcg/kg/**hour**; admixed in combination with bupivacaine or ropivacaine (Ref).

Chronic cancer pain; breakthrough

Chronic cancer pain; breakthrough:

Note: For use in patients who are tolerant to and currently receiving opioid therapy for persistent cancer pain; dosing should be individually titrated to provide adequate analgesia with minimal side effects. **Patients must remain on around-the-clock opioids during use.**

Transmucosal lozenge (eg, Actiq): **Note:** Fentanyl lozenge (brand and generic) has been discontinued in the United States >1 year. Do **not** convert patients from any other fentanyl product to Actiq on a mcg-per-mcg basis. Patients previously using another fentanyl product should be initiated in usual manner; individually titrate to provide adequate analgesia while minimizing adverse effects.

Adolescents ≥ 16 years:

Initial: 200 mcg (consumed over 15 minutes) for all patients; if after 30 minutes from the start of the lozenge (ie, 15 minutes following the completion of the lozenge), the pain is unrelieved, a second 200 mcg dose may be given over 15 minutes. A maximum of 1 additional dose can be given per pain episode; **must wait at least 4 hours before treating another episode.** To limit the number of units in the home during titration, only prescribe an initial titration supply of six 200 mcg lozenges.

Dose titration: Dose titration should be done if patient requires more than 1 dose/breakthrough pain episode for several consecutive episodes. From the initial dose, closely follow patients and modify the dose until patient reaches a dose providing adequate analgesia using a single dosage unit per breakthrough cancer pain episode. If signs/symptoms of excessive opioid effects (eg, respiratory depression) occur, immediately remove the dosage unit from the patient's mouth, dispose of properly, and reduce subsequent doses. If adequate relief is not achieved 15 minutes after completion of the first dose (ie, 30 minutes after the start of the lozenge), only 1 additional lozenge of the same strength may be given for that episode; **must wait at least 4 hours before treating another episode.**

Maintenance dose: Once titrated to an effective dose, patients should generally use a single dosage unit per breakthrough pain episode. During any pain episode, if adequate relief is not achieved 15 minutes after completion of the first dose (ie, 30 minutes after the start of the lozenge), only 1 additional lozenge of the same strength may be given over 15 minutes for that episode; **must wait at least 4 hours before treating another episode.** Consumption should be limited to ≤ 4 units per day (once an effective breakthrough dose is found). If adequate analgesia is **not** provided after treating several episodes of breakthrough pain using the same dose, increase dose to next highest lozenge strength (initially dispense no more than 6 units of the new strength). Consider increasing the around-the-clock opioid therapy in patients experiencing >4 breakthrough pain episodes per day and have their long-term opioid re-evaluated. If signs/symptoms of excessive opioid effects (eg, respiratory depression) occur, immediately remove the dosage unit from the patient's mouth, dispose of properly, and reduce subsequent doses.

Chronic pain, moderate to severe

Chronic pain, moderate to severe (opioid-tolerant):

Transdermal patch (eg, Duragesic): **Note:** For use in patients who are opioid-tolerant receiving at least 60 mg oral morphine equivalents per day. Discontinue or taper all other around-the-clock or extended-release opioids when initiating therapy with fentanyl transdermal patch:

Children ≥ 2 years and Adolescents:

Initial: 25 mcg/hour system or higher based on previous opioid dosing. To convert patients from oral or parenteral opioids to transdermal patch, a 24-hour analgesic requirement should be calculated (based on prior opioid use). Using the following tables, the appropriate initial dose can be determined. The initial fentanyl dosage may be approximated from the 24-hour morphine dosage equivalent and titrated to minimize adverse effects and provide analgesia. Substantial interpatient variability exists in relative potency. Therefore, it is safer to underestimate a patient's daily fentanyl requirement and provide breakthrough pain relief with rescue medication (eg, immediate release opioid) than to overestimate requirements. With the initial application, the absorption of transdermal fentanyl requires several hours to reach plateau; therefore, transdermal fentanyl is inappropriate for management of acute pain. Change patch every 72 hours.

Conversion from continuous infusion of fentanyl: In patients who have adequate pain relief with a fentanyl infusion, fentanyl may be converted to transdermal dosing at a rate equivalent to the intravenous rate. Based on experience in adults, a two-step taper of the infusion to be completed over 12 hours may be considered (Ref) after the patch is applied. The infusion is decreased to 50%

of the original rate 6 hours after the application of the first patch, and subsequently discontinued twelve hours after application.

Titration: Short-acting agents may be required until analgesic efficacy is established and/or as supplements for “breakthrough” pain. The amount of supplemental doses should be closely monitored. Appropriate dosage increases may be based on daily supplemental dosage using the ratio of 45 mg/24 hours of oral morphine to a 12.5 mcg/hour increase in fentanyl dosage.

Frequency of adjustment: The dosage should not be titrated more frequently than every 3 days after the initial dose or every 6 days thereafter. Titrate dose based on the daily dose of supplemental opioids required by the patient on the second or third day of the initial application. **Note:** Upon discontinuation, ~17 hours are required for a 50% decrease in fentanyl levels.

Frequency of application: The majority of patients may be controlled on every 72-hour administration; however, a small number of adult patients have required every 48-hour administration.

Discontinuation: When discontinuing transdermal fentanyl and not converting to another opioid, use a gradual downward titration, such as decreasing the dose by 50% every 6 days, to reduce the possibility of withdrawal symptoms.

Dose conversion guidelines for transdermal fentanyl from other opioids (see tables).

Note: The conversion factors in these tables are only to be used for the conversion from current opioid therapy to Duragesic patch. Conversion factors in this table cannot be used to convert from Duragesic to another opioid (doing so may lead to fatal overdose due to overestimation of the new opioid). US and Canadian dose conversion guidelines differ; consult table for US recommendations. The Canadian product is not approved in pediatric patients. **These are not tables of equianalgesic doses.**

US Labeling: Dose Conversion Guidelines: Recommended Initial Duragesic Dose Based Upon Daily Oral Morphine Dose^a

Oral 24-Hour Morphine

(mg/day)

Duragesic Dose^{b,c}

(mcg/hour)

^a **The table should NOT be used to convert from transdermal fentanyl (Duragesic) to other opioid analgesics.** Rather, following removal of the patch, titrate the dose of the new opioid until adequate analgesia is achieved.

^b Pediatric patients initiating therapy on a 25 mcg/hour Duragesic system should be opioid-tolerant and receiving at least 60 mg oral morphine equivalents per day.

^c A fentanyl 37.5 mcg/hour transdermal system is also available and may be considered during conversion from prior opioids or dose titration.

60 to 134 mg/day

25 mcg/hour

135 to 224 mg/day

50 mcg/hour

225 to 314 mg/day

75 mcg/hour

315 to 404 mg/day

100 mcg/hour

405 to 494 mg/day

125 mcg/hour
495 to 584 mg/day
150 mcg/hour
585 to 674 mg/day
175 mcg/hour
675 to 764 mg/day
200 mcg/hour
765 to 854 mg/day
225 mcg/hour
855 to 944 mg/day
250 mcg/hour
945 to 1,034 mg/day
275 mcg/hour
1,035 to 1,124 mg/day
300 mcg/hour

US Labeling: Dose Conversion Guidelines^a

Current Analgesic
Total Daily Dosage
(mg/day)

^a **The table should NOT be used to convert from transdermal fentanyl (Duragesic) to other opioid analgesics.** Rather, following removal of the patch, titrate the dose of the new opioid until adequate analgesia is achieved.

Morphine (IM/IV)
10 to 22 mg/day
23 to 37 mg/day
38 to 52 mg/day
53 to 67 mg/day
Oxycodone (oral)
30 to 67 mg/day
67.5 to 112 mg/day
112.5 to 157 mg/day
157.5 to 202 mg/day
Codeine (oral)
150 to 447 mg/day
-
-
-
Hydromorphone (oral)

8 to 17 mg/day
17.1 to 28 mg/day
28.1 to 39 mg/day
39.1 to 51 mg/day
Hydromorphone (IV)
1.5 to 3.4 mg/day
3.5 to 5.6 mg/day
5.7 to 7.9 mg/day
8 to 10 mg/day
Meperidine (IM)
75 to 165 mg/day
166 to 278 mg/day
279 to 390 mg/day
391 to 503 mg/day
Methadone (oral)
20 to 44 mg/day
45 to 74 mg/day
75 to 104 mg/day
105 to 134 mg/day

Duragesic (fentanyl transdermal) recommended dose (mcg/hour)

25 mcg/hour

50 mcg/hour

75 mcg/hour

100 mcg/hour

Pain and sedation, patients who are critically ill

Pain and sedation, patients who are critically ill (ICU):

Infants and Children: Limited data available in <2 years of age: Initial IV bolus: 1 to 2 mcg/kg followed by continuous IV infusion at initial rate: 1 mcg/kg/**hour**; titrate to effect; usual range: 1 to 3 mcg/kg/**hour**; some patients may require higher rates (5 mcg/kg/**hour**) (Ref).

Adolescents:

Patient weight ≤50 kg: Initial IV bolus: 0.5 to 2 mcg/kg followed by continuous IV infusion at initial rate: 0.5 to 2 mcg/kg/**hour** based on expert recommendations for children and pediatric patients ≤50 kg (Ref).

Patient weight >50 kg: Initial IV bolus: 25 to 100 mcg/dose followed by continuous IV infusion at initial rate: 25 to 200 mcg/**hour** based on expert recommendations for pediatric patients and experience in adult patients (Ref).

Pain, patient-controlled analgesia

Pain, patient-controlled analgesia (PCA): Limited data available:

Note: PCA has been used in children as young as 5 years; however, clinicians need to assess children <7 years of age to determine if they are able to use the PCA device correctly. All patients should

receive an initial loading dose of an analgesic (to attain adequate control of pain) before starting PCA for maintenance. Adjust doses, lockouts, and limits based on required loading dose, age, state of health, and presence of opioid tolerance. Use lower end of dosing range for opioid-naïve. Assess patient and pain control at regular intervals and adjust settings if needed (Ref). Concentrations of PCA should be standardized, refer to institutional policy. The following are general dosing suggestions; refer to institutional policy.

Children ≥ 5 years and Adolescents:

Patient weight ≤ 50 kg:

Demand dose: Usual initial range: IV: 0.2 to 0.5 mcg/kg/dose; an initial dose of 0.3 mcg/kg/dose has also been suggested (Ref).

Lockout interval: Range: 5 to 15 minutes (Ref).

Usual basal rate (optional): IV: 0 to 0.5 mcg/kg/**hour** (Ref); an alternative range of 0.15 to 1 mcg/kg/**hour** (with an initial of 0.2 mcg/kg/**hour**) has also been suggested (Ref).

Maximum dose: Format of maximum total amount of fentanyl per unit of time varies with pump settings, ordering standards, and institutional policy. The following are different options that have been reported; refer to institutional policy.

1-hour dose limit: Initial: 5 demand doses/**hour** (Ref).

2-hour dose limit: 80% of maximum quantity based on the sum of the basal rate and demand doses allowed in a 2-hour period of time (Ref).

4-hour dose limit: 7 to 10 mcg/kg (Ref).

Patient weight > 50 kg (Ref):

Loading dose: 25 mcg.

Demand dose: Usual initial: 5 to 25 mcg.

Lockout interval: Usual: 8 to 10 minutes.

Procedural pain and sedation

Procedural pain and sedation : Limited data available:

Note : Dose and route of administration may vary with procedure; refer to institutional policy.

Injection:

Infants and Children: IV: 1 to 2 mcg/kg/dose; administer 3 minutes before the procedure; maximum dose: 50 mcg/dose; may repeat $\frac{1}{2}$ original dose every 3 to 5 minutes if necessary; titrate to effect (Ref).

Adolescents: IV: 0.5 to 1 mcg/kg/dose; may repeat after 1 to 2 minutes (Ref); or 25 to 50 mcg, repeat full dose in 5 minutes if needed, may repeat 4 to 5 times with 25 mcg at 5-minute intervals if needed (Ref).

Intranasal (using parenteral preparation): Infants and Children weighing ≥ 10 kg and Adolescents: Intranasal: 1.5 to 2 mcg/kg administered 15 minutes prior to procedure; maximum dose: 100 mcg/dose (Ref).

Rapid sequence intubation, pretreatment

Rapid sequence intubation, pretreatment: Limited data available: Infants and Children: IV, Intraosseous: Usual dose: 1 mcg/kg/dose, reported range: 1 to 2 mcg/kg/dose; may repeat every 3 minutes; maximum dose: 50 to 100 mcg/dose (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Injection: There are no dosage adjustments provided in the manufacturer's labeling; however, the following guidelines have been used by some clinicians (Ref):

Infants, Children, and Adolescents: The following assumes dosages of 0.5 to 2 mcg/kg/dose or 1 to 5 mcg/kg/**hour** in normal renal function: IV:

GFR >50 mL/minute/1.73 m²: No adjustment required.

GFR 10 to 50 mL/minute/1.73 m²: Administer 75% of usual dose.

GFR <10 mL/minute/1.73 m²: Administer 50% of usual dose.

Intermittent hemodialysis: Administer 50% of usual dose.

Peritoneal dialysis (PD): Administer 50% of usual dose.

Continuous renal replacement therapy (CRRT): Administer 75% of usual dose.

Transdermal (patch): Pediatric patients ≥2 years: Degree of impairment (ie, CrCl) not defined in manufacturer's labeling.

Mild to moderate impairment: Initial: Reduce dose by 50%.

Severe impairment: Use not recommended.

Transmucosal (lozenge [eg, Actiq]) and nasal spray: Adolescents ≥16 years: Although fentanyl pharmacokinetics may be altered in renal disease, fentanyl can be used successfully in the management of breakthrough cancer pain. Use with caution; reduce initial dose and titrate to reach clinical effect with careful monitoring of patients, especially those with severe renal disease.

Dosing: Liver Impairment: Pediatric

Injection: There are no dosage adjustments provided in the manufacturer's labeling.

Transdermal (patch): Pediatric patients ≥2 years:

Mild to moderate impairment: Initial: Reduce dose by 50%.

Severe impairment: Use not recommended.

Transmucosal (lozenge [eg, Actiq]): Adolescents ≥16 years: Although fentanyl pharmacokinetics may be altered in hepatic disease, fentanyl can be used successfully in the management of breakthrough cancer pain. Use with caution; reduce initial dose and titrate to reach clinical effect with careful monitoring of patients, especially those with severe hepatic disease.

Adverse Reactions (Significant): Considerations

Opioid-induced constipation

Opioid-induced **constipation** (OIC) is the most common subtype of opioid-induced bowel dysfunction (OIBD), which is a broader term that encompasses additional GI opioid-induced adverse reactions including nausea, vomiting, and gastroesophageal reflux. Tolerance does not develop to OIC (Ref). Constipation due to fentanyl may decrease quality of life, contribute to nonadherence, and result in treatment failure (Ref). Transdermal fentanyl may cause less constipation than oral opioids (eg, oxycodone, morphine) (Ref). Fentanyl may cause less constipation than morphine in critically ill children (Ref).

Mechanism: Time-related; activity in both the central and enteric nervous systems, predominantly via the mu-opioid receptor, delays gastric emptying, inhibits peristalsis, impairs enzyme release, and produces antisecretory effects (Ref).

Onset: Varied; changes in peristalsis may occur 5 to 25 minutes after administration of opioids (Ref). However, OIC is defined based on a 7-day period of change (Ref).

Risk factors:

- Chronic opioid administration (Ref)

Opioid-induced respiratory depression

Serious, life-threatening, or fatal opioid-induced **respiratory depression** (OIRD) may occur with use of fentanyl (Ref). Fentanyl may be particularly hazardous secondary to potency, faster onset of action, and lower cross tolerance (Ref).

Mechanism: Dose-related; mediated predominately via the mu-opioid receptor. Activates specific CNS sites (ie, pre-Bötzinger complex and Kölliker-Fuse nucleus) that control respiratory rhythms (Ref).

Onset: Rapid; OIRD reported from 5 minutes to 1.2 hours (Ref).

Risk factors:

- Overdose (but may also occur at therapeutic doses)
- Initiation or dose escalation
- Opioid naive (Ref)
- Opioid misuse/abuse (Ref)
- Respiratory disease (eg, obstructive sleep apnea) (Ref)
- Cardiac disease (Ref)
- Morbid obesity (Ref)
- Older adults
- Kidney and/or liver impairment
- Cachectic or debilitated
- Concurrent benzodiazepines or other CNS depressants
- Concurrent use with cytochrome P450 3A4 inhibitors

Opioid-induced withdrawal

Abrupt discontinuation or dose reduction in patients who are physically dependent (ie, have **opioid dependence**) to opioids has been associated with a serious opioid-induced **withdrawal syndrome** (OIW), uncontrolled pain, attempts to find other opioids (including illicit), and suicide in all ages. Concurrent use of mixed agonist/antagonist analgesics (eg, pentazocine, nalbuphine, butorphanol) or partial agonist (eg, buprenorphine) analgesics may also precipitate withdrawal symptoms and/or reduced analgesic efficacy in patients following prolonged therapy with mu opioid agonists. Reversal of analgesia, irritability, nausea, and vomiting, abdominal cramping, tachycardia, restlessness, and sweating are common symptoms. Symptoms typically dissipate over 4 to 7 days and fully resolve within 14 days; may cause significant distress but are rarely life-threatening (Ref).

Mechanism: Withdrawal; abrupt discontinuation in patients who are physically dependent on opioids results in noradrenergic hyperactivity. This hyperactivity occurs due to abrupt reversal of adaptive mechanisms related to cAMP, locus coeruleus firing rate, and norepinephrine levels that occur with prolonged exposure to opioids (Ref).

Onset: Rapid; symptoms typically occur within 12 hours after discontinuation and peak at 36 to 72 hours; varies based on half-life of opioid (Ref).

Risk factors:

- Prolonged exposure to opioids (Ref)
- Pediatrics: Neonates >1 week, infants, and children <22 months: Duration of ≥ 9 days or a total dose ≥ 2.5 mg/kg has been associated with a 100% chance of developing opioid withdrawal; similarly, a duration of 5 days or a total dose of 1.5 mg/kg has a reported 50% chance of developing opioid withdrawal (Ref)
- Opioid use disorder (Ref)
- Abrupt discontinuation or dose reduction (Ref)

- Concurrent use of mixed agonist/antagonist analgesics (eg, pentazocine, nalbuphine, butorphanol) or partial agonist (eg, buprenorphine) analgesics
- Discontinuation of concurrent cytochrome P450 3A4 inhibitors

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%:

Cardiovascular: Peripheral edema (5% to 32%)

Endocrine & metabolic: Dehydration (9% to 21%), hypokalemia (2% to 15%), weight loss (3% to 11%)

Gastrointestinal: Abdominal pain (3% to 15%), anorexia (5% to 11%), constipation (1% to 26%), diarrhea (6% to 16%), nausea (2% to 42%), vomiting (1% to 37%)

Hematologic & oncologic: Anemia (9% to 32%)

Nervous system: Asthenia (2% to 30%), confusion (1% to 16%), depression (1% to 11%), dizziness (1% to 32%), drowsiness (3% to 20%), fatigue (2% to 20%), headache (2% to 17%), insomnia (1% to 11%)

Neuromuscular & skeletal: Back pain (4% to 11%)

Respiratory: Dyspnea (2% to 19%), pneumonia (2% to 16%)

1% to 10%:

Cardiovascular: Bradycardia ($\geq 1\%$), chest pain ($\geq 1\%$), chest wall pain ($\geq 1\%$), deep vein thrombosis ($\geq 1\%$), edema ($\geq 1\%$), hypertension ($\geq 1\%$), hypotension (2%), palpitations (4%), pulmonary embolism (nasal spray: $\geq 1\%$), sinus tachycardia ($\geq 1\%$), tachycardia ($\geq 1\%$), vascular injury ($\geq 1\%$), vasodilation (1% to 4%)

Dermatologic: Alopecia ($\geq 1\%$), cellulitis ($\geq 1\%$), dermal ulcer ($\geq 1\%$), diaphoresis (1% to 3%), ecchymoses ($\geq 1\%$), erythema of skin (1%), hyperhidrosis ($\leq 6\%$), pallor ($\geq 1\%$), pressure ulcer ($\geq 1\%$), pruritus (1% to 5%), skin rash (1% to 8%)

Endocrine & metabolic: Cachexia ($\geq 1\%$), hypercalcemia ($\geq 1\%$), hyperglycemia ($\geq 1\%$), hypoalbuminemia ($\geq 1\%$), hypocalcemia ($\geq 1\%$), hypomagnesemia ($\geq 1\%$), hyponatremia ($\geq 1\%$), lactic acidosis ($\geq 1\%$)

Gastrointestinal: Abdominal distention ($\geq 1\%$), decreased appetite ($\geq 1\%$), dysgeusia ($\geq 1\%$), dyspepsia ($\geq 1\%$), dysphagia (lozenge, buccal tablet, sublingual spray: $\geq 1\%$), eructation ($\geq 1\%$), flatulence ($\geq 1\%$), gastritis ($\geq 1\%$), gastroenteritis ($\geq 1\%$), gastroesophageal reflux disease ($\geq 1\%$), gastrointestinal hemorrhage ($\geq 1\%$), gastrointestinal signs and symptoms (reduced oral intake; buccal tablet: $\geq 1\%$), gingival pain (buccal tablet: $\geq 1\%$), gingivitis (lozenge: $\geq 1\%$), glossalgia ($\geq 1\%$), glossitis (lozenge: $\geq 1\%$), hematemesis ($\geq 1\%$), intestinal obstruction (2% to 4%; subileus: $< 1\%$), mucosal swelling (buccal tablet: $\geq 1\%$), oral candidiasis (lozenge, buccal tablet, sublingual spray: $\geq 1\%$), oral herpes simplex infection ($\geq 1\%$), oral mucosa ulcer (lozenge, buccal tablet, nasal spray: $\geq 1\%$; including lip ulceration), periodontal abscess (lozenge: $\geq 1\%$), rectal disease ($\geq 1\%$), rectal hemorrhage ($\geq 1\%$), rectal pain ($\geq 1\%$), stomach discomfort ($\geq 1\%$), stomatitis (lozenge, buccal tablet, sublingual spray: $\geq 1\%$), tooth abscess (buccal tablet: $\geq 1\%$), upper abdominal pain (3%), xerostomia (1% to 4%)

Genitourinary: Breast neoplasm ($\geq 1\%$), difficulty in micturition ($\geq 1\%$), dysuria ($\geq 1\%$), hematuria ($\geq 1\%$), mastalgia ($\geq 1\%$), pelvic pain ($\geq 1\%$), scrotal edema ($\geq 1\%$), urinary incontinence ($\geq 1\%$), urinary retention (1% to 3%), urinary tract infection ($\geq 1\%$), urinary urgency ($\geq 1\%$), vaginal hemorrhage ($\geq 1\%$), vaginitis ($\geq 1\%$)

Hematologic & oncologic: Bruise ($\geq 1\%$), leukopenia ($\geq 1\%$), lymphadenopathy ($\geq 1\%$), lymphedema ($\geq 1\%$), neutropenia (2% to 8%), pancytopenia ($\geq 1\%$), thrombocytopenia ($\geq 1\%$)

Hepatic: Ascites ($\geq 1\%$), increased serum alkaline phosphatase ($\geq 1\%$), increased serum aspartate aminotransferase ($\geq 1\%$), jaundice ($\geq 1\%$)

Hypersensitivity: Hypersensitivity reaction ($\geq 1\%$)

Infection: Infection ($\geq 1\%$; including fungal infection, viral infection), influenza ($\geq 1\%$), sepsis ($\geq 1\%$)

Local: Application-site irritation (buccal tablet, sublingual spray: $\leq 3\%$), application-site pain (buccal tablet: 4%), application-site reaction (buccal tablet, transdermal patch: $\leq 10\%$), application-site vesicles (including ulcer: buccal tablet, lozenge, transdermal patch: 3%)

Nervous system: Abnormal dreams (1%), abnormal gait (1% to 5%), agitation ($\geq 1\%$), altered sense of smell ($\geq 1\%$), amnesia ($\geq 1\%$), anxiety (2% to 9%), balance impairment ($\geq 1\%$), changes in thinking (1% to 2%), chills ($\geq 1\%$), disorientation ($\geq 1\%$), emotional lability ($\geq 1\%$), euphoria ($\geq 1\%$), falling ($\geq 1\%$), hallucination (1% to 2%), hypertonia (2%), hypoesthesia ($\geq 1\%$), irritability ($\geq 1\%$), lethargy ($\geq 1\%$), malaise (4%), migraine ($\geq 1\%$), myasthenia ($\geq 1\%$), myoclonus (4%), nervousness (1% to 4%), neuropathy ($\geq 1\%$), opioid withdrawal syndrome ($\geq 1\%$), paresthesia (2%), peripheral neuropathy ($\geq 1\%$), restlessness ($\geq 1\%$), sedated state ($\geq 1\%$), seizure (1% to 2%), sensation of cold (6%), speech disturbance ($\geq 1\%$), stupor (1% to 4%), tremor (1% to 3%), vertigo (1% to 4%)

Neuromuscular & skeletal: Arthralgia (3% to 8%), arthropathy ($\geq 1\%$), bone disease ($\geq 1\%$), hypokinesia ($\geq 1\%$), limb pain ($\geq 1\%$), lower limb cramp ($\geq 1\%$), muscle spasm (4%), myalgia ($\geq 1\%$), neck pain ($\geq 1\%$), ostealgia ($\geq 1\%$), pathological fracture ($\geq 1\%$; including vertebral compression fracture), shoulder pain ($\geq 1\%$)

Ophthalmic: Blepharoptosis (nasal spray: $\geq 1\%$), blurred vision ($\geq 1\%$), conjunctivitis ($\geq 1\%$), dry eye syndrome (nasal spray: $\geq 1\%$), strabismus (nasal spray: $\geq 1\%$), swelling of eye (nasal spray: $\geq 1\%$), visual disturbance (1% to 2%)

Otic: Ear disease ($\geq 1\%$; including otalgia), tinnitus ($\geq 1\%$)

Renal: Hydronephrosis ($\geq 1\%$), kidney failure ($\geq 1\%$)

Respiratory: Abnormal breath sounds (decreased: $\geq 1\%$), asthma ($\geq 1\%$), bradypnea ($\geq 1\%$), bronchitis ($\geq 1\%$), cough (3% to 9%; increased cough: $\geq 1\%$), dyspnea on exertion ($\geq 1\%$), epistaxis ($\geq 1\%$), flu-like symptoms ($\geq 1\%$), hemoptysis ($\geq 1\%$), hypoxia ($\geq 1\%$), increased bronchial secretions ($\geq 1\%$), laryngitis (sublingual spray: $\geq 1\%$), nasal congestion (nasal spray: $\geq 1\%$), nasal discomfort (nasal spray: $\geq 1\%$), nasopharyngitis ($\geq 1\%$), pharyngitis ($\geq 1\%$), pharyngolaryngeal pain ($\geq 1\%$), pleural effusion ($\geq 1\%$), post nasal drip (nasal spray: $\geq 1\%$), rhinitis ($\geq 1\%$), rhinorrhea (nasal spray: $\geq 1\%$), sinusitis ($\geq 1\%$), upper respiratory tract infection ($\geq 1\%$), wheezing ($\geq 1\%$)

Miscellaneous: Fever (5% to 7%)

<1%:

Cardiovascular: Angina pectoris, orthostatic hypotension, peripheral vascular disease

Dermatologic: Allergic dermatitis, cheilitis, contact dermatitis, contact hypersensitivity (transdermal patch, application site), eczema, exfoliative dermatitis, maculopapular rash, skin discoloration, urticaria, vesiculobullous rash (lozenge)

Endocrine & metabolic: Acidosis, decreased libido, hot flash, hypoglycemia, increased thirst

Gastrointestinal: Dental caries (lozenge), esophagitis, fecal impaction, fecal incontinence, gingival hemorrhage (lozenge), hiccups, mucous membrane disease

Genitourinary: Erectile dysfunction, nocturia, oliguria, polyuria, sexual disorder

Hematologic & oncologic: Hemorrhage, hypoproteinemia, prolonged bleeding time

Hepatic: Hepatorenal syndrome, liver tenderness

Hypersensitivity: Facial edema

Infection: Bacterial infection, herpes zoster infection

Local: Application-site dermatitis (transdermal patch)

Nervous system: Ataxia, cerebral ischemia, delirium, facial nerve paralysis, hemiplegia, subdural hematoma, voice disorder

Neuromuscular & skeletal: Amyotrophy, arthritis, foot-drop, muscle twitching, myopathy, neck stiffness, synovitis, tendinopathy

Ophthalmic: Disease of the lacrimal apparatus, injury to eye region (hemorrhage), miosis

Otic: Deafness, reversible hearing loss

Renal: Flank pain, pyelonephritis, renal pain

Respiratory: Cyanosis, hyperventilation, pneumothorax, pulmonary disease, respiratory depression, respiratory failure, respiratory insufficiency

Miscellaneous: Cyst, granuloma

Frequency not defined:

Cardiovascular: Flushing

Nervous system: Drug abuse, neonatal withdrawal, opioid dependence

Postmarketing:

Dermatologic: Skin erosion (application-site, transdermal patch)

Gastrointestinal: Gingival recession (lozenge), tooth loss (lozenge)

Genitourinary: Hypogonadism (Brennan 2013; Debono 2011)

Hypersensitivity: Anaphylaxis

Nervous system: Allodynia (opioid-induced hyperalgesia) (FDA Safety Communication 2023), impaired consciousness, loss of consciousness

Neuromuscular & skeletal: Laryngospasm, muscle rigidity (including chest wall rigidity) (Rosal 2021)

Respiratory: Apnea, hypoventilation, respiratory distress

Contraindications

Hypersensitivity (eg, anaphylaxis, hypersensitivity) to fentanyl or any component of the formulation.

Additional contraindications for transdermal patch: Significant respiratory depression; acute or severe bronchial asthma in an unmonitored setting or in the absence of resuscitative equipment; gastrointestinal obstruction, including paralytic ileus (known or suspected); patients requiring short-term therapy, management of acute or intermittent pain, postoperative or mild pain; patients who are **not** opioid tolerant.

Additional contraindications for transmucosal buccal tablets (Fentora), lozenges (Actiq), sublingual spray (Subsys), intranasal (Lazanda): Significant respiratory depression (Actiq, Fentora only); acute or severe bronchial asthma in an unmonitored setting or in the absence of resuscitative equipment; gastrointestinal obstruction, including paralytic ileus (known or suspected); acute or postoperative pain (including headache, migraine, or dental pain); patients who are **not** opioid tolerant; acute pain management in the emergency room.

Canadian labeling: Additional contraindications (not in US labeling):

Injection: Hypersensitivity to other opioids; known or suspected mechanical GI obstruction (eg, bowel obstruction, strictures) or any diseases/conditions that affect bowel transit (eg, ileus of any type); suspected surgical abdomen (eg, acute appendicitis, pancreatitis); mild pain that can be managed with other pain medications; acute or severe bronchial asthma, chronic obstructive airway, or status asthmaticus; acute respiratory depression; hypercapnia; cor pulmonale; acute alcoholism, delirium tremens, and convulsive disorders; severe CNS depression, increased cerebrospinal or intracranial pressure and head injury; concurrent use or use within 14 days of a monoamine oxidase inhibitor

(MAOI); septicemia; severe hemorrhage or shock; local infection at proposed injection site; disturbances in blood morphology and/or anticoagulant therapy or other concomitant drug therapy or medical conditions which could contraindicate the technique of epidural administration.

Transdermal patch: Hypersensitivity to other opioids; suspected surgical abdomen (eg, acute appendicitis, pancreatitis); known or suspected mechanical GI obstruction (eg, bowel obstruction, strictures) or any diseases/conditions that affect bowel transit (eg, ileus of any type); acute alcoholism, delirium tremens, and convulsive disorders; severe CNS depression, increased cerebrospinal or intracranial pressure and head injury; concurrent use of MAOIs or within 14 days of therapy; perioperative pain; women who are nursing, pregnant, or during labor and delivery.

Transmucosal buccal tablets (Fentora): Hypersensitivity to other opioids; acute pain management in the emergency room; known or suspected mechanical GI obstruction (eg, bowel obstruction or strictures) or any diseases/conditions that affect bowel transit (eg, ileus of any type); suspected surgical abdomen (eg, acute appendicitis or pancreatitis); acute or severe bronchial asthma, chronic obstructive airway, status asthmaticus; acute respiratory depression; hypercapnia; cor pulmonale; acute alcoholism, delirium tremens, and convulsive disorders; severe CNS depression, increased cerebrospinal or intracranial pressure and head injury; concurrent use or use within 14 days of an MAOI.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Warnings/Precautions

Concerns related to adverse effects:

- Esophageal dysfunction: Opioid-induced esophageal dysfunction has been reported; risk increases with higher doses and prolonged therapy.
- Hyperalgesia: Opioid-induced hyperalgesia (OIH) has occurred with short-term and prolonged use of opioid analgesics. Symptoms may include increased levels of pain upon opioid dosage increase, decreased levels of pain upon opioid dosage decrease, or pain from ordinarily nonpainful stimuli; symptoms may be suggestive of OIH if there is no evidence of underlying disease progression, opioid tolerance, opioid withdrawal, or addictive behavior. Consider decreasing the current opioid dose or opioid rotation in patients who experience OIH.
- Hypotension: May cause severe hypotension (including orthostatic hypotension and syncope); use with caution in patients with hypovolemia, cardiovascular disease (including acute MI), or drugs which may exaggerate hypotensive effects (including phenothiazines or general anesthetics). Monitor for symptoms of hypotension following initiation or dose titration. Avoid use in patients with circulatory shock.
- Serotonin syndrome: Potentially life-threatening serotonin syndrome (SS) has occurred with concomitant use of fentanyl and serotonergic agents (eg, selective serotonin reuptake inhibitors, serotonin and norepinephrine reuptake inhibitors, triptans, tricyclic antidepressants, 5-HT₃ receptor antagonists, mirtazapine, trazodone, tramadol) and agents that impair metabolism of serotonin (eg, monoamine oxidase inhibitors). Monitor patients closely for signs of SS such as mental status changes (eg, agitation, hallucinations, delirium, coma); autonomic instability (eg, tachycardia, labile BP, diaphoresis); neuromuscular changes (eg, tremor, rigidity, myoclonus); and GI symptoms (eg, nausea, vomiting, diarrhea). Discontinue fentanyl if SS is suspected.

Disease-related concerns:

- Abdominal conditions: May obscure diagnosis or clinical course of patients with acute abdominal conditions.
- Adrenocortical insufficiency: Use with caution in patients with adrenal insufficiency, including Addison's disease. Long-term opioid use may cause secondary hypogonadism, which may lead to mood disorders and osteoporosis (Brennan 2013).
- Allergic rhinitis: Intranasal: Allergic rhinitis is not expected to alter fentanyl absorption following nasal administration; however, use of nasal decongestants (eg, oxymetazoline) during episodes of

rhinitis may result in lower peak concentrations and delayed $T_{\max}$, therefore, titration of intranasal fentanyl is not recommended during use of nasal decongestants.

- Biliary tract impairment: Use with caution in patients with biliary tract dysfunction, including acute pancreatitis; may cause constriction of sphincter of Oddi.
- Bradycardia: Use with caution in patients with bradycardia or bradyarrhythmias (may produce further bradycardia).
- CNS depression/coma: Avoid use in patients with impaired consciousness or coma as these patients are susceptible to intracranial effects of CO_2 retention.
- Delirium tremens: Use with caution in patients with delirium tremens.
- Head trauma: Use with extreme caution in patients with head injury, intracranial lesions, or elevated intracranial pressure (ICP); exaggerated elevation of ICP may occur.
- Mental health conditions: Use opioids with caution for chronic pain in patients with mental health conditions (eg, depression, anxiety disorders, post-traumatic stress disorder) due to potential increased risk for opioid use disorder and overdose; more frequent monitoring is recommended (CDC [Dowell 2022]).
- Obesity: Use with caution in patients who are morbidly obese.
- Oral mucositis: Sublingual spray (Subsys): Cancer patients with oral mucositis experienced increased fentanyl exposure following sublingual spray administration; avoid use in patients with $\geq$ grade 2 mucositis; use with caution in patients with grade 1 mucositis, and closely monitor for respiratory and CNS depression.
- Prostatic hyperplasia/urinary stricture: Use with caution in patients with prostatic hyperplasia and/or urinary stricture.
- Psychosis: Use with caution in patients with toxic psychosis.
- Renal impairment: Use with caution in patients with renal impairment.
- Respiratory disease: Use with caution and monitor for respiratory depression in patients with significant chronic obstructive pulmonary disease or cor pulmonale, and those with a substantially decreased respiratory reserve, hypoxia, hypercapnia, or preexisting respiratory depression, particularly when initiating and titrating therapy; critical respiratory depression may occur, even at therapeutic dosages. Consider the use of alternative nonopioid analgesics in these patients.
- Seizures: Use with caution in patients with a history of seizure disorders; may cause or exacerbate seizures.
- Sleep-related disorders: Use with caution in patients with sleep-related disorders, including sleep apnea, due to increased risk for respiratory and central nervous system depression. Monitor carefully and titrate dosage cautiously in patients with mild sleep-disordered breathing. Avoid opioids in patients with moderate to severe sleep-disordered breathing (CDC [Dowell 2022]).
- Thyroid dysfunction: Use with caution in patients with thyroid dysfunction.

Special populations:

- Older adult: Use opioids with caution in older adults; may be more sensitive to adverse effects. Clearance may also be reduced in older adults (with or without renal impairment) resulting in a narrow therapeutic window and increased adverse effects. Monitor closely for adverse effects associated with opioid therapy (eg, respiratory and central nervous system depression, falls, cognitive impairment, constipation) (CDC [Dowell 2022]). Consider the use of alternative nonopioid analgesics in these patients when possible.

Dosage form specific issues:

- Transdermal patch:

- Serum fentanyl concentrations may increase by approximately one-third for patients with a body temperature of 40°C (104°F) secondary to a temperature-dependent increase in fentanyl release from the patch and increased skin permeability.
- Avoid unclothed/unwashed application site exposure, inadvertent person-to-person patch transfer (eg, while hugging), incidental exposure (eg, sharing same bed, sitting on patch), intentional exposure (eg, chewing), or accidental exposure by caregivers when applying/removing patch.
- May contain conducting metal (eg, aluminum); remove patch prior to MRI.

Other warnings/precautions:

- Abuse/misuse/diversion: Use with caution in patients with a history of substance abuse disorder; potential for drug dependency exists. Other factors associated with increased risk for misuse include concomitant depression or other mental health conditions, higher opioid dosages, long-term use, or taking other CNS depressants. Consider offering an opioid overdose reversal agent (eg, naloxone, nalmefene) prescription in patients with an increased risk for overdose, such as history of overdose or substance use disorder, higher opioid dosages (≥ 50 morphine milligram equivalents [MME]/day orally), concomitant benzodiazepine use, and patients at risk for returning to a high dose after losing tolerance (CDC [Dowell 2022]).

- Appropriate use: Outpatient setting: Opioids should **not** be used as first-line therapy for acute (<1-month duration), subacute (1- to 3-month duration), or chronic pain (>3-month duration [outside of end-of-life or palliative care, active cancer treatment, sickle cell disease, or medication-based opioid use disorder treatment]). Preferred management includes nonpharmacologic therapy and nonopioid therapy (eg, nonsteroidal anti-inflammatory drugs, acetaminophen, certain antiseizure medications and antidepressants) as appropriate for the specific condition. If opioid therapy is initiated, it should be combined with nonpharmacologic and non-opioid therapy, as appropriate. Prior to initiation, known risks and realistic benefits of opioid therapy should be discussed with the patient. Therapy should be initiated at the lowest effective dosage using IR opioids (instead of ER/long-acting opioids). For the treatment of acute pain, therapy should only be given for the expected duration of pain severe enough to require opioids and prescribed as needed (not scheduled). For the treatment of subacute and chronic pain, realistic treatment goals for pain/function should be established, including consideration for discontinuation if benefits do not outweigh risks. Therapy should be continued only if clinically meaningful improvement in pain/function outweighs risks. Risk to patients increases with higher opioid dosages. Dosages ≥ 50 MME/day are likely to not have increased benefit to pain relief or function relative to overall risk to patients; before increasing dosage to ≥ 50 MME/day, readdress pain and reassess evidence of individual benefits and risks (CDC [Dowell 2022]).

- Opioid overdose reversal agent access: Discuss the availability of opioid overdose reversal agents (eg, naloxone, nalmefene) with all patients who are prescribed opioid analgesics, as well as their caregivers, and consider prescribing it to patients who are at increased risk of opioid overdose. These include patients who are also taking benzodiazepines or other CNS depressants, have an opioid use disorder (OUD) (current or history of) or substance use disorder, have higher opioid dosages (≥ 50 MME/day orally), patients at risk for returning to a high dose after losing tolerance (CDC [Dowell 2022]), or have experienced opioid-induced respiratory depression/opioid overdose. Additionally, health care providers should consider prescribing an opioid overdose reversal agent to patients prescribed medications to treat OUD; patients at risk of opioid overdose even if they are not taking an opioid analgesic or medication to treat OUD; and patients taking opioids, including methadone or buprenorphine for OUD, if they have household members, including children, or other close contacts at risk for accidental ingestion or opioid overdose. Inform patients and caregivers on options for obtaining opioid overdose reversal agents (eg, by prescription, OTC, directly from a pharmacist, a community-based program) as permitted by state dispensing and prescribing guidelines. Educate patients and caregivers on how to recognize respiratory depression, proper administration, important differences of opioid overdose reversal agents, and getting emergency help, even if an opioid overdose reversal agent is administered.

- Optimal regimen: An opioid-containing analgesic regimen should be tailored to each patient's needs and based upon the type of pain being treated (acute versus chronic), the route of administration, degree of tolerance for opioids (naive versus chronic user), age, weight, and medical con-

dition. The optimal analgesic dose varies widely among patients; doses should be titrated to pain relief/prevention.

- Surgery: Opioids decrease bowel motility; monitor for decreased bowel motility in postoperative patients receiving opioids. Use with caution in the perioperative setting; individualize treatment when transitioning from parenteral to oral analgesics.

Warnings: Additional Pediatric Considerations

Dosage form specific:

Use transdermal patch in pediatric patients only if they are opioid-tolerant, receiving at least 60 mg oral morphine equivalents per day, and ≥ 2 years of age.

Use of Actiq was evaluated in a clinical trial of 15 opioid-tolerant pediatric patients (age: 5 to 15 years) with breakthrough pain; 12 of the 15 patients received doses of 200 mcg to 600 mcg; no conclusions about safety and efficacy could be drawn due to the small sample size.

Some dosage forms may contain propylene glycol; in neonates large amounts of propylene glycol delivered orally, intravenously (eg, $>3,000$ mg/day), or topically have been associated with potentially fatal toxicities which can include metabolic acidosis, seizures, renal failure, and CNS depression; toxicities have also been reported in children and adults including hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Shehab 2009).

Product Availability

Fentanyl buccal tablets (brand and generic), Fentanyl lozenge on a handle (brand and generic), Lazanda (intranasal solution) and Subsys (sublingual spray) have been discontinued in the United States for >1 year.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Liquid, Sublingual, as base:

Subsys: 100 mcg (10 ea [DSC]); 200 mcg (30 ea [DSC]); 400 mcg (30 ea [DSC]); 600 mcg (30 ea [DSC]); 800 mcg (1 ea [DSC], 30 ea [DSC]); 1200 mcg (2 ea [DSC]); 1600 mcg (2 ea [DSC]) [contains alcohol, usp, levomenthol, propylene glycol]

Lozenge on a Handle, Buccal, as citrate [strength expressed as base]:

Actiq: 200 mcg (1 ea [DSC], 30 ea [DSC]); 400 mcg (1 ea [DSC], 30 ea [DSC]); 600 mcg (1 ea [DSC], 30 ea [DSC]); 800 mcg (1 ea [DSC], 30 ea [DSC]); 1200 mcg (1 ea [DSC], 30 ea [DSC]); 1600 mcg (1 ea [DSC], 30 ea [DSC]) [berry flavor]

Generic: 200 mcg (1 ea [DSC], 30 ea [DSC]); 400 mcg (1 ea [DSC], 30 ea [DSC]); 600 mcg (1 ea [DSC], 30 ea [DSC]); 800 mcg (1 ea [DSC], 30 ea [DSC]); 1200 mcg (1 ea [DSC], 30 ea [DSC]); 1600 mcg (1 ea [DSC], 30 ea [DSC])

Patch 72 Hour, Transdermal, as base:

Generic: 100 [delivers 100 mcg/hr] (1 ea, 5 ea); 12 [delivers 12.5 mcg/hr] (1 ea, 5 ea); 25 [delivers 25 mcg/hr] (1 ea, 5 ea); 37.5 [delivers 37.5 mcg/hr] (1 ea, 5 ea); 50 [delivers 50 mcg/hr] (1 ea, 5 ea); 62.5 [delivers 62.5 mcg/hr] (1 ea, 5 ea); 75 [delivers 75 mcg/hr] (1 ea, 5 ea); 87.5 [delivers 87.5 mcg/hr] (1 ea, 5 ea)

Solution, Injection [preservative free]:

Generic: 2500 mcg/50 mL (50 mL)

Solution, Injection, as citrate [preservative free]:

Generic: 50 mcg/mL (1 mL); 2500 mcg/50 mL (50 mL)

Solution, Injection, as citrate [strength expressed as base, preservative free]:

Generic: 100 mcg/2 mL (2 mL); 1000 mcg/20 mL (20 mL); 250 mcg/5 mL (5 mL); 500 mcg/10 mL (10 mL)

Solution, Intravenous, as citrate [strength expressed as base]:

Generic: 2500 mcg/50 mL (50 mL [DSC])

Solution, Nasal, as citrate [strength expressed as base]:

Lazanda: 100 mcg/actuation (1 ea [DSC]); 400 mcg/actuation (1 ea [DSC]) [contains propylparaben (propyl p-hydroxybenzoate)]

Solution Prefilled Syringe, Injection, as citrate:

Generic: 100 mcg/2 mL (2 mL [DSC])

Solution Prefilled Syringe, Injection, as citrate [preservative free]:

Generic: 25 mcg/0.5 mL (0.5 mL); 50 mcg/mL (1 mL); 100 mcg/2 mL (2 mL)

Tablet, Buccal, as citrate [strength expressed as base]:

Fentora: 100 mcg [DSC], 200 mcg [DSC], 400 mcg [DSC], 600 mcg [DSC], 800 mcg [DSC]

Generic: 100 mcg [DSC], 200 mcg [DSC], 400 mcg [DSC], 600 mcg [DSC], 800 mcg [DSC]

Generic Equivalent Available: US

May be product dependent

Pricing: US

Patch, 72-hour (fentaNYL Transdermal)

12 mcg/hr (per each): \$20.30

25 mcg/hr (per each): \$14.42 - \$21.27

37.5 mcg/hr (per each): \$65.31

50 mcg/hr (per each): \$26.36 - \$38.88

62.5 mcg/hr (per each): \$94.91

75 mcg/hr (per each): \$40.21 - \$59.31

87.5 mcg/hr (per each): \$129.22

100 mcg/hr (per each): \$53.36 - \$78.71

Solution (fentaNYL Citrate (PF) Injection)

50 mcg/mL (per mL): \$2.34 - \$2.65

100 mcg/2 mL (per mL): \$0.86 - \$2.51

250 mcg/5 mL (per mL): \$0.36 - \$0.82

500MCG/10ML (per mL): \$1.22

1000 mcg/20 mL (per mL): \$0.51 - \$0.66

2500 mcg/50 mL (per mL): \$0.38 - \$1.12

Solution Prefilled Syringe (fentaNYL Citrate (PF) Injection)

25 mcg/0.5 mL (per 0.5 mL): \$2.65 - \$2.65

50 mcg/mL (per mL): \$2.65 - \$3.31

Solution Prefilled Syringe (fentaNYL Citrate Injection)

100 mcg/2 mL (per mL): \$3.12

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Patch 72 Hour, Transdermal:

Generic: 37 mcg/hr (5 ea)

Patch 72 Hour, Transdermal, as base:

Duragesic 50: 50 [delivers 50 mcg/hr] ([DSC])

Generic: 100 [delivers 100 mcg/hr] (5 ea); 12 [delivers 12.5 mcg/hr] (5 ea); 25 [delivers 25 mcg/hr] (5 ea); 50 [delivers 50 mcg/hr] (5 ea); 75 [delivers 75 mcg/hr] (5 ea)

Solution, Injection, as citrate:

Generic: 50 mcg/mL (1 mL, 2 mL, 5 mL, 10 mL, 20 mL, 50 mL)

Solution, Injection, as citrate [strength expressed as base]:

Generic: 100 mcg/2 mL (2 mL)

Tablet, Buccal, as citrate [strength expressed as base]:

Fentora: 100 mcg, 200 mcg, 400 mcg, 600 mcg, 800 mcg

Controlled Substance

C-II

Administration: Adult

Epidural (Canadian labeling; not in US labeling): For postoperative pain management may administer as bolus dose (diluted in preservative free NS to a final concentration of 10 mcg/mL) or by continuous infusion at a rate of 1 mcg/kg/hour. Use within 24 hours.

Parenteral: Administer IM or IV slowly over 1 to 2 minutes; rapid IV infusion may result in skeletal muscle and chest wall rigidity, impaired ventilation, or respiratory distress/arrest. May administer as a continuous IV infusion or patient-controlled analgesia (off label). May also administer SUBQ (intermittent or continuous infusion [off label]).

Transdermal patch: Apply to nonirritated and nonirradiated skin on a flat surface, such as chest, back, flank, or upper arm; apply to the upper back in patients with cognitive impairment. Hair at application site may be clipped (do not shave). If application site must be cleaned prior to application, clean site with clear water and allow to dry completely. Do not use damaged, cut or leaking patches; patch may be less effective. Do not use soaps, oils, lotions, alcohol, or any other agents to cleanse skin before application because they may irritate the skin or alter its characteristics. Immediately after removal from sealed package, firmly press patch in place and hold for 30 seconds. Wash hands immediately with soap and water after applying patch. Change patch every 72 hours; apply new patch to a different skin site. Contact with unwashed or unclothed application sites can result in secondary exposure to fentanyl and should be avoided. Avoid exposing application site to external heat sources (eg, heating pad, electric blanket, heat lamp, tanning lamps, hot tub, sunbathing, saunas, heated water beds). If there is difficulty with patch adhesion, the edges of the system may be taped in

place with first-aid tape. If there is continued difficulty with adhesion, an adhesive film dressing (eg, Bioclusive, Tegaderm) may be applied over the system. Dispose of any used or unused patches by removing patch from protective pouch and liner, fold adhesive ends together and flush patch down toilet immediately. Do not flush pouch or protective liner as such items can be discarded in the trash. For used and unused patches, the Canadian labeling recommends folding adhesive ends together and returning to a pharmacy for proper disposal; temporary storage in a biohazard container may be used before returning to pharmacy.

Lozenge: Foil overwrap should be removed just prior to administration. Place the unit in mouth between the cheek and gum and allow it to dissolve. Do not chew. Lozenge may be moved from one side of the mouth to the other using the handle. The unit should be consumed over a period of 15 minutes. After consumption of a complete unit, the handle may be disposed of in a trash container that is out of the reach of children. For a unit that still has any drug matrix remaining on the handle, the handle should be placed under hot running tap water until the entire drug matrix has dissolved. Special child-resistant containers are available to temporarily store partially consumed units that cannot be disposed of immediately. Refer to manufacturer's labeling for additional disposal instructions.

Buccal tablet: Patient should not open blister until ready to administer. The blister backing should be peeled back to expose the tablet; tablet should not be pushed out through the blister. Immediately use tablet once removed from blister. Place entire tablet in the buccal cavity (above a rear molar, between the upper cheek and gum) or under the tongue (US labeling recommends for maintenance dosing only; Canadian labeling does not restrict sublingual use to maintenance dosing only); should dissolve in about 14 to 25 minutes. If remnants remain after 30 minutes, they may be swallowed with water. Tablet should not be split, crushed, sucked, chewed, or swallowed whole. When possible, alternate sides of mouth with each dose.

Intranasal:

Lazanda: Prior to initial use, prime device by spraying 4 sprays into the provided pouch (the counting window will show a green bar when the bottle is ready for use). Insert nozzle a short distance into the nose (~1/2 inch or 1 cm) and point towards the bridge of the nose (while closing off the other nostril using 1 finger). Press on finger grips until a "click" sound is heard and the number in the counting window advances by one. The "click" sound and dose counter are the only reliable methods for ensuring a dose has been administered (spray is not always felt on the nasal mucosa). Patient should remain seated for at least 1 minute following administration. Do not blow nose for ≥30 minutes after administration. Wash hands before and after use. If not used within 5 days, re-prime by spraying once. There are 8 full therapeutic sprays in each bottle; do not continue to use bottle after "8" sprays have been used. Dispose of bottle and contents if it has been ≥60 days since first use. Before disposal, all unopened or partially used bottles must be completely emptied by spraying the contents into the provided pouch. After "8" therapeutic sprays has been reached on the counter, patients should continue to spray an additional four sprays into the pouch to ensure that any residual fentanyl has been expelled (an audible click will no longer be heard and the counter will not advance beyond "8"). The empty bottle and the sealed pouch must be put into the child-resistant container before placing in the trash. Wash hands with soap and water immediately after handling the pouch. If the pouch is lost, another one can be ordered by the patient or caregiver by calling 1-866-458-6389.

Parenteral solution (off-label route): Using 50 mcg/mL parenteral solution, administer half the dose in each nostril using a mucosal atomizer device (preferred) or drip into nostril with a needleless syringe; maximum volume per nostril: 1 mL (Ref).

SUBQ continuous infusions: Change site every 3 to 7 days or when erythema occurs (Ref). To maintain comfort, the SUBQ infusion rate should generally not exceed 5 **mL**/hour (Ref).

Sublingual spray: Open sealed blister unit with scissors immediately prior to administration. Contents of unit should be sprayed into mouth under the tongue. Dispose of each unit dose immediately after use; place used unit into one of the provided white disposal bags. After sealing appropriately, discard in the trash. Dispose of any unused units as soon as no longer needed. Prior to disposal, empty all the medicine into the provided charcoal-lined disposal pouch. The disposal pouch should then be placed into the white disposal bag, sealed appropriately, and discarded in the trash.

Preparation for Administration: Adult

Continuous IV infusion: May further dilute in NS or D5W

Usual Infusion Concentrations: Adult

IV infusion: 10 mcg/mL

Administration: Pediatric

Parenteral: IV:

Neonates: Administer as a continuous infusion via an infusion pump or by slow IV push over 3 to 5 minutes. Larger bolus doses (>5 mcg/kg) should be administered over 5 to 10 minutes; may also administer intermittent doses via syringe pump over 15 to 30 minutes (Ref). Risk of chest wall rigidity is minimized when doses are administered over at least 1 to 2 minutes (Ref).

Infants, Children, and Adolescents: Administer by slow IV push over 3 to 5 minutes or by continuous infusion via an infusion pump. Larger bolus doses (>5 mcg/kg) should be given by slow IV push over 5 to 10 minutes.

Transdermal patch: Children ≥ 2 years and Adolescents: Apply to nonhairy, clean, dry, nonirritated, intact skin of the flat area of front or back of upper torso, flank area, or upper arm; apply to upper back in young children or in people with cognitive impairment to decrease the potential of the patient removing the patch. Monitor the adhesion of the system closely in children. Clip hair prior to application, do **not** shave area; prior to application, skin may be cleaned with clear water (do not use soaps, lotions, alcohol, oils, or other substances which may irritate the skin); allow skin to dry thoroughly prior to application. Apply patch immediately after removing from package; firmly press in place and hold for at least 30 seconds; change patch every 72 hours; remove old patch before applying new patch; do not apply new patch to same place as old patch; wash hands after applying patch. If there is difficulty with patch adhesion, the edges of the system may be taped in place with first-aid tape; if difficulty with adhesion persists, an adhesive film dressing (eg, Bioclusive, Tegaderm) may be applied over the system. If patch falls off before 72 hours, a new patch may be applied to a different skin site. Dispose of any used or unused patches by removing patch from protective pouch and liner, fold adhesive ends together and flush patch down toilet immediately. Do not flush pouch or protective liner as such items can be discarded in the trash.

Note: Transdermal patch is a membrane-controlled system; do **not** cut the patch to deliver partial doses; do **not** use patches that are cut, damaged, or leaking; do **not** use if seal of package is broken; rate of drug delivery may be significantly increased if patch is cut, damaged, or leaking and result in absorption of a potentially fatal dose; reservoir contents and adhesion may be affected if cut; if partial dose is needed, surface area of patch can be blocked proportionally using adhesive bandage (Ref). Do **not** use soap, alcohol, or other solvents to remove transdermal gel if it accidentally touches skin as they may increase transdermal absorption; use copious amounts of water. Avoid exposing application site to external heat sources (eg, electric blanket, heating pad, heat lamp, tanning lamp, sauna, heated water bed, hot tub, hot baths, sunbathing). Dispose of properly.

Transmucosal products:

Intranasal (using parenteral preparation):

Solution (parenteral preparation): If congested, suction nostrils prior to administration. Using a 50 mcg/mL solution, administer half of the dose to each nostril using an atomizer such as the MAD Nasal Drug delivery device or drip into the nostril slowly with a syringe; higher concentrations (150 mcg/mL and 300 mcg/mL) have been studied to prevent excess volume administration, but comparative trials have not been performed (Ref).

Oral lozenge (eg, Actiq): Adolescents ≥ 16 years: Oral: Foil overwrap should be removed just prior to administration, patient should place the unit in mouth between cheek and lower gum; occasionally move lozenge from one side of the mouth to the other and allow it to dissolve. Do not bite or chew lozenge; consume lozenge over 15 minutes. Remove handle after lozenge is consumed. Early removal should be considered if the patient has achieved an adequate response and/or shows signs of respiratory depression. After consumption of a complete unit, the handle may be disposed of in a

trash container that is out of the reach of children. For a partially consumed unit, or a unit that still has any drug matrix remaining on the handle, the handle should be placed under hot running tap water until the drug matrix has dissolved. Special child-resistant containers are available to temporarily store partially consumed units that cannot be disposed of immediately.

Preparation for Administration: Pediatric

Parenteral: Intermittent IV doses or continuous IV infusion: May further dilute in NS or D5W. Vermont Oxford Network recommends standard concentrations of 2 mcg/mL or 10 mcg/mL for both intermittent doses and continuous infusions for neonates (Ref); in pediatric patients, ASHP recommends standard concentrations of 10 mcg/mL and 50 mcg/mL (undiluted) for continuous infusions (Ref).

Usual Infusion Concentrations: Pediatric

IV infusion: 10 mcg/mL, 50 mcg/mL.

Storage/Stability

Injection: Store intact vials in original carton at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Protect from light.

Intranasal: Do not store above 25°C (77°F); do not freeze. Protect from light. Bottle should be stored in the provided child-resistant container when not in use and kept out of the reach of children at all times.

Transdermal patch, transmucosal (buccal tablet, lozenge, sublingual spray): Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Protect transmucosal products from freezing and moisture. Keep all products out of the reach of children.

Use: Labeled Indications

Pain management, acute and chronic pain:

Injection: Surgery: Adjunct to general or regional anesthesia; preoperative medication; analgesic during anesthesia and in the immediate postoperative period.

Transdermal patch: Chronic pain: Management of severe and persistent pain in opioid-tolerant patients that requires an opioid analgesic and cannot be adequately treated with alternative options, including IR opioids.

Limitations of use: Because of the risks of substance use disorder, abuse, misuse, overdose, and death, which may occur at any dosage or duration and persist over the course of therapy, reserve opioid analgesics, including fentanyl, for use in patients for whom alternative treatment options are ineffective, not tolerated, or would be otherwise inadequate to provide sufficient management of pain. Not indicated as an as-needed (prn) analgesic.

Transmucosal lozenge (eg, Actiq), buccal tablet (Fentora), intranasal (Lazanda), sublingual spray (Subsys): Cancer pain, breakthrough: Management of breakthrough cancer pain in opioid-tolerant patients ≥ 18 years (Fentora, Lazanda, Subsys) and ≥ 16 years (Actiq) of age who are already receiving and who are tolerant to around-the-clock opioid therapy for their underlying persistent cancer pain.

Limitations of use: Not for use in opioid non-tolerant patients. Not for use in the management of acute or postoperative pain, including headache/migraine, dental pain, or in the emergency department. As a part of the TIRF REMS Access program, these products may be dispensed only to outpatients enrolled in the program. For inpatient administration (eg, hospitals, hospices, and long-term care facilities that prescribe for inpatient use), patient and prescriber enrollment is not required.

Note: "Opioid-tolerant" patients are defined as patients who are taking at least:

Oral morphine 60 mg/day, **or**

Transdermal fentanyl 25 mcg/hour, **or**

Oral oxycodone 30 mg/day, **or**

Oral hydromorphone 8 mg/day, **or**

Oral oxymorphone 25 mg/day, **or**

Oral hydrocodone 60 mg/day, **or**

Equianalgesic dose of another opioid for at least 1 week

Use: Off-Label: Adult

Critically ill patients in the ICU, analgesia and sedation; Procedural sedation and analgesia, outside the operating room; Rapid sequence intubation, pretreatment

Medication Safety Issues

Sound-alike/look-alike issues:

FentaNYL may be confused with ALfentanil, SUFentanil

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (anesthetic agent, general, inhaled and IV; epidural and intrathecal medications; opioids, all formulations and routes of administration) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024; ISMP List of High-Alert Medications in Community/Ambulatory Care Settings ©ISMP 2021; High-Alert Medications in Long-Term Care (LTC) Settings ©ISMP 2021).

Older Adult: High-Risk Medication:

Fentanyl is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥ 65 years of age) due to an increased risk of exacerbation of constipation. Opioids are not recommended as first-line treatment of mild pain or long-term treatment of osteoarthritis. Short-acting opioids should be available for breakthrough pain in patients on long-acting opioids. Avoid use in patients with a history of recurrent falls (O'Mahony 2023).

Administration issues:

Fentanyl transdermal system patches: Leakage of fentanyl gel from the patch has been reported; patch may be less effective; do not use. Thoroughly wash any skin surfaces coming into direct contact with gel with water (do not use soap). May contain conducting metal (eg, aluminum); remove patch prior to MRI.

Other safety concerns:

Fentanyl transdermal system patches:

Avoid use of patches in opioid-naïve patients and/or acute pain. Patches are only intended for use in opioid-tolerant patients; steps should be taken to verify opioid status and type of pain being treated prior to prescribing to prevent inappropriate use. Serious harm, including fatalities, are associated with use of patches in opioid-naïve patients. Dosing of transdermal fentanyl patches may be confusing. Transdermal fentanyl patches should always be prescribed in mcg/hour, not size. Patch dosage form labeled as 12 mcg/hour actually delivers 12.5 mcg/hour of fentanyl. Use caution, as orders may be written as "Fentanyl 12.5" which can be erroneously interpreted as a 125 mcg dose.

Patches should be stored and disposed of with care to avoid accidental exposure to children. The FDA has issued numerous safety advisories to warn users of the possible consequences (including hospitalization and death) of inappropriate storage or disposal of patches. According to the 2020 to 2021 *ISMP Targeted Medication Safety Best Practices for Hospitals*, patches should **not** be stored in automated dispensing cabinets or as unit stock in clinical areas where acute pain is primarily treated (eg, emergency departments, operating rooms, postanesthesia care units, procedural areas).

Actiq, Fentora, and Subsys are not interchangeable; do not substitute doses on a mcg-per-mcg basis.

Metabolism/Transport Effects

Substrate of CYP3A4 (Major); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Agents with Clinically Relevant Anticholinergic Effects: May increase adverse/toxic effects of Opioid Agonists. Specifically, the risk for constipation and urinary retention may be increased with this combination. *Risk C: Monitor*

Alcohol (Ethyl): May increase CNS depressant effects of Opioid Agonists. *Risk X: Avoid*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

Almotriptan: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Alosetron: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Alvimopan: Opioid Agonists may increase adverse/toxic effects of Alvimopan. This is most notable for patients receiving long-term (i.e., more than 7 days) opiates prior to alvimopan initiation. Management: Alvimopan is contraindicated in patients receiving therapeutic doses of opioids for more than 7 consecutive days immediately prior to alvimopan initiation. *Risk D: Consider Therapy Modification*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amphetamines: May increase analgesic effects of Opioid Agonists. *Risk C: Monitor*

Amphetamines: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability). Initiate amphetamines at lower doses, monitor frequently, and adjust doses as needed. *Risk C: Monitor*

Antiemetics (5HT3 Antagonists): May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Antiplatelet Agents (P2Y12 Inhibitors): FentaNYL may decrease antiplatelet effects of Antiplatelet Agents (P2Y12 Inhibitors). FentaNYL may decrease serum concentrations of Antiplatelet Agents (P2Y12 Inhibitors). *Risk C: Monitor*

Antipsychotic Agents: Serotonergic Agents (High Risk) may increase adverse/toxic effects of Antipsychotic Agents. Specifically, serotonergic agents may enhance dopamine blockade, possibly increasing the risk for neuroleptic malignant syndrome. Antipsychotic Agents may increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bradycardia-Causing Agents: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

Buprenorphine: May decrease therapeutic effects of Opioid Agonists. Management: Seek alternatives to buprenorphine in patients receiving pure opioid agonists. If combined in certain pain management situations (eg, surgery), monitor for symptoms of therapeutic failure/high dose requirements or opioid withdrawal symptoms. *Risk D: Consider Therapy Modification*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

BusPIRone: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Ceritinib: FentaNYL may increase bradycardic effects of Ceritinib. Ceritinib may increase serum concentrations of FentaNYL. Management: Consider fentanyl dose reductions when combined with ceritinib. Monitor for respiratory depression, sedation, and bradycardia. Upon discontinuation of ceritinib, consider a fentanyl dose increase; monitor for signs and symptoms of withdrawal. *Risk D: Consider Therapy Modification*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and chlorpromazine when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Cimetidine: May increase analgesic effects of Opioid Agonists. Cimetidine may increase respiratory depressant effects of Opioid Agonists. *Risk C: Monitor*

Clofazimine: May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). *Risk C: Monitor*

CNS Depressants: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

CYP3A4 Inducers (Moderate): May decrease serum concentrations of FentaNYL. *Risk C: Monitor*

CYP3A4 Inducers (Strong): May decrease serum concentrations of FentaNYL. *Risk C: Monitor*

CYP3A4 Inhibitors (Moderate): May increase serum concentrations of FentaNYL. Management: Consider fentanyl dose reductions when combined with a moderate CYP3A4 inhibitor. Monitor for respiratory depression and sedation. Upon discontinuation of a CYP3A4 inhibitor, consider a fentanyl dose increase; monitor for signs and symptoms of withdrawal. *Risk D: Consider Therapy Modification*

CYP3A4 Inhibitors (Strong): May increase serum concentrations of FentaNYL. Management: Consider fentanyl dose reductions when combined with a strong CYP3A4 inhibitor. Monitor for respiratory depression and sedation. Upon discontinuation of a CYP3A4 inhibitor, consider a fentanyl dose increase; monitor for signs and symptoms of withdrawal. *Risk D: Consider Therapy Modification*

Dapoxetine: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Do not use serotonergic agents (high risk) with dapoxetine or within 7 days of serotonergic agent discontinuation. Do not use dapoxetine within 14 days of monoamine oxidase inhibitor use. Dapoxetine labeling lists this combination as contraindicated. *Risk X: Avoid*

Decongestants: May decrease serum concentrations of FentaNYL. *Risk C: Monitor*

Desmopressin: Opioid Agonists may increase hyponatremic effects of Desmopressin. *Risk C: Monitor*

Dexmethylphenidate-Methylphenidate: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Dextromethorphan: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Diuretics: Opioid Agonists may increase adverse/toxic effects of Diuretics. Opioid Agonists may decrease therapeutic effects of Diuretics. *Risk C: Monitor*

Eletriptan: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Eluxadolone: Opioid Agonists may increase constipating effects of Eluxadolone. *Risk X: Avoid*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ergot Derivatives: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Etrasimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Fexinidazole: Bradycardia-Causing Agents may increase arrhythmogenic effects of Fexinidazole. *Risk X: Avoid*

Fingolimod: Bradycardia-Causing Agents may increase bradycardic effects of Fingolimod. Management: Consult with the prescriber of any bradycardia-causing agent to see if the agent could be switched to an agent that does not cause bradycardia prior to initiating fingolimod. If combined, perform continuous ECG monitoring after the first fingolimod dose. *Risk D: Consider Therapy Modification*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Fusidic Acid (Systemic): May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). Management: Consider avoiding this combination if possible. If required, monitor patients closely for increased adverse effects of the CYP3A4 substrate. *Risk D: Consider Therapy Modification*

Gastrointestinal Agents (Prokinetic): Opioid Agonists may decrease therapeutic effects of Gastrointestinal Agents (Prokinetic). *Risk C: Monitor*

Gepirone: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. *Risk C: Monitor*

Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies): May decrease therapeutic effects of Opioid Agonists. Opioid Agonists may decrease therapeutic effects of Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies). *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Isocarboxazid: CNS Depressants may increase adverse/toxic effects of Isocarboxazid. *Risk X: Avoid*

Ivabradine: Bradycardia-Causing Agents may increase bradycardic effects of Ivabradine. *Risk C: Monitor*

Ixabepilone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Landiolol: Bradycardia-Causing Agents may increase bradycardic effects of Landiolol. *Risk X: Avoid*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Linezolid: FentaNYL may increase serotonergic effects of Linezolid. This could result in serotonin syndrome. Management: Consider alternatives to the coadministration of fentanyl and linezolid when possible, as product labeling states the concomitant use of fentanyl with MAOIs such as linezolid, or use of such as agents within 14 days is not recommended. *Risk D: Consider Therapy Modification*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Meperidine: May increase serotonergic effects of FentaNYL. This could result in serotonin syndrome. Meperidine may increase CNS depressant effects of FentaNYL. Management: Consider alternatives to this combination. If use is necessary, monitor for signs and symptoms of serotonin syndrome/serotonin toxicity and CNS depression. *Risk D: Consider Therapy Modification*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methotrimeprazine: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and methotrimeprazine when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Methylene Blue: May increase serotonergic effects of FentaNYL. This could result in serotonin syndrome. Management: Consider alternatives to the coadministration of fentanyl and methylene blue when possible, as product labeling states the concomitant use of fentanyl is not recommended for patients taking MAOIs or within 14 days of stopping such treatment. *Risk D: Consider Therapy Modification*

Metoclopramide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metoclopramide: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Consider monitoring for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

Midodrine: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

MiFEPRIStone: May increase serum concentrations of FentaNYL. *Risk X: Avoid*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Monoamine Oxidase Inhibitors (Antidepressant): FentaNYL may increase serotonergic effects of Monoamine Oxidase Inhibitors (Antidepressant). This could result in serotonin syndrome. Management: Consider alternatives when possible as the use of fentanyl with antidepressant monoamine oxidase inhibitors (MAOIs), or within 14 days of stopping MAOI therapy, is not recommended. This is listed as a contraindication in some non-US labels. *Risk D: Consider Therapy Modification*

Monoamine Oxidase Inhibitors (Type B): FentaNYL may increase serotonergic effects of Monoamine Oxidase Inhibitors (Type B). This could result in serotonin syndrome. Management: Consider alternatives when possible, as the use of fentanyl with type B monoamine oxidase inhibitors (MAOIs), or within 14 days of stopping MAOI therapy, is not recommended. This is listed as a contraindication in some non-US labels. *Risk D: Consider Therapy Modification*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Nalfurafine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nalfurafine: Opioid Agonists may increase adverse/toxic effects of Nalfurafine. Opioid Agonists may decrease therapeutic effects of Nalfurafine. *Risk C: Monitor*

Nalmefene: May decrease therapeutic effects of Opioid Agonists. Management: Avoid the concomitant use of oral nalmefene and opioid agonists. Discontinue oral nalmefene 1 week prior to any anticipated use of opioid agonists. If combined, larger doses of opioid agonists will likely be required. *Risk D: Consider Therapy Modification*

Naltrexone: May increase or decrease therapeutic effects of Opioid Agonists. Management: Seek therapeutic alternatives to opioids. See full drug interaction monograph for detailed recommendations. *Risk X: Avoid*

Nefazodone: FentaNYL may increase serotonergic effects of Nefazodone. This could result in serotonin syndrome. Nefazodone may increase serum concentrations of FentaNYL. Management: Consider reducing fentanyl dose. Monitor for signs and symptoms of respiratory depression, sedation, and serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia) when these agents are combined. *Risk D: Consider Therapy Modification*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Ondansetron: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opioids (Mixed Agonist / Antagonist): May decrease analgesic effects of Opioid Agonists. Management: Seek alternatives to mixed agonist/antagonist opioids in patients receiving pure opioid agonists, and monitor for symptoms of therapeutic failure/high dose requirements (or withdrawal in opioid-dependent patients) if patients receive these combinations. *Risk X: Avoid*

Opipramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opipramol: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Oxitriptan: Serotonergic Agents (High Risk) may increase serotonergic effects of Oxitriptan. This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Ozanimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Pegvisomant: Opioid Agonists may decrease therapeutic effects of Pegvisomant. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

PHENobarbital: May increase CNS depressant effects of FentaNYL. PHENobarbital may decrease serum concentrations of FentaNYL. Management: Avoid use of fentanyl and phenobarbital when possible. Monitor for respiratory depression/sedation. Because phenobarbital is also a moderate CYP3A4 inducer, monitor for decreased fentanyl efficacy and withdrawal if combined. *Risk D: Consider Therapy Modification*

Pipamperone: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Ponesimod: Bradycardia-Causing Agents may increase bradycardic effects of Ponesimod. Management: Avoid coadministration of ponesimod with drugs that may cause bradycardia when possible. If combined, monitor heart rate closely and consider obtaining a cardiology consult. Do not initiate ponesimod in patients on beta-blockers if HR is less than 55 bpm. *Risk D: Consider Therapy Modification*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Primidone: May increase CNS depressant effects of FentaNYL. Primidone may decrease serum concentrations of FentaNYL. Management: Avoid use of fentanyl and primidone when possible. Monitor for respiratory depression/sedation. Because primidone is also a moderate CYP3A4 inducer, monitor for decreased fentanyl efficacy and withdrawal if combined. *Risk D: Consider Therapy Modification*

Procarbazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Propofol: May increase CNS depressant effects of FentaNYL. Management: Consider alternatives to this combination when possible. If the combination is used, monitor more closely for bradycardia, apnea, and excessive CNS depression. Propofol induction dose requirements may be reduced. Pediatric patients may be at greater risk. *Risk D: Consider Therapy Modification*

Ramosetron: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Ramosetron: Opioid Agonists may increase constipating effects of Ramosetron. *Risk C: Monitor*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

ROPINIrole: CNS Depressants may increase sedative effects of ROPINIrole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Samidorphan: May decrease therapeutic effects of Opioid Agonists. Specifically, the risk for opioid withdrawal may be increased. *Risk X: Avoid*

Selective Serotonin Reuptake Inhibitor: Serotonergic Opioids (High Risk) may increase serotonergic effects of Selective Serotonin Reuptake Inhibitor. This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) if these agents are combined. *Risk C: Monitor*

Serotonergic Agents (High Risk, Miscellaneous): Serotonergic Opioids (High Risk) may increase serotonergic effects of Serotonergic Agents (High Risk, Miscellaneous). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) if these agents are combined. *Risk C: Monitor*

Serotonergic Non-Opioid CNS Depressants: May increase serotonergic effects of Serotonergic Opioids (High Risk). This could result in serotonin syndrome. Serotonergic Non-Opioid CNS Depressants may increase CNS depressant effects of Serotonergic Opioids (High Risk). Management: Consider alternatives to this drug combination. If combined, monitor for signs and symptoms of serotonin syndrome/serotonin toxicity and CNS depression. *Risk D: Consider Therapy Modification*

Serotonin 5-HT_{1D} Receptor Agonists (Triptans): May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Serotonin/Norepinephrine Reuptake Inhibitor: FentaNYL may increase serotonergic effects of Serotonin/Norepinephrine Reuptake Inhibitor. This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) if these agents are combined. *Risk C: Monitor*

Sincalide: Drugs that Affect Gallbladder Function may decrease therapeutic effects of Sincalide. Management: Consider discontinuing drugs that may affect gallbladder motility prior to the use of sincalide to stimulate gallbladder contraction. *Risk D: Consider Therapy Modification*

Siponimod: Bradycardia-Causing Agents may increase bradycardic effects of Siponimod. Management: Avoid coadministration of siponimod with drugs that may cause bradycardia. If combined, consider obtaining a cardiology consult regarding patient monitoring. *Risk D: Consider Therapy Modification*

Somatostatin Analogs: May increase or decrease analgesic effects of Opioid Agonists. *Risk C: Monitor*

St John's Wort: May increase serotonergic effects of FentaNYL. This could result in serotonin syndrome. St John's Wort may decrease serum concentrations of FentaNYL. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity and reduced fentanyl effects (including withdrawal symptoms) when combined. Monitor for increased fentanyl effects if St. John's wort is discontinued. *Risk C: Monitor*

Succinylcholine: May increase bradycardic effects of Opioid Agonists. *Risk C: Monitor*

Sulpiride: Serotonergic Agents (High Risk) may increase adverse/toxic effects of Sulpiride. Specifically, serotonergic agents may enhance dopamine blockade, possibly increasing the risk for neuroleptic malignant syndrome. Sulpiride may increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. *Risk C: Monitor*

Syrian Rue: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Tedizolid: May increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Consider monitoring for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

Tilidine: May increase therapeutic effects of Opioid Agonists. *Risk X: Avoid*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

TraMADol: Serotonergic Opioids (High Risk) may increase serotonergic effects of TraMADol. This could result in serotonin syndrome. Serotonergic Opioids (High Risk) may increase CNS depressant effects of TraMADol. Management: Consider alternatives to this drug combination. If combined, monitor for signs and symptoms of serotonin syndrome/serotonin toxicity and CNS depression. *Risk D: Consider Therapy Modification*

Tricyclic Antidepressants: Serotonergic Opioids (High Risk) may increase serotonergic effects of Tricyclic Antidepressants. This could result in serotonin syndrome. Tricyclic Antidepressants may increase CNS depressant effects of Serotonergic Opioids (High Risk). Management: Consider alternatives to this drug combination. If combined, monitor for signs and symptoms of serotonin syndrome/serotonin toxicity and CNS depression. *Risk D: Consider Therapy Modification*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Zolpidem: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and zolpidem when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Reproductive Considerations

Chronic opioid use may cause hypogonadism and hyperprolactinemia which may decrease fertility in patients of reproductive potential. Menstrual cycle disorders (including amenorrhea), erectile dysfunction, and impotence have been reported. The incidence of hypogonadism may be increased with the use of opioids in high doses or long-acting opioid formulations. It is not known if the effects on fertility are reversible. Monitor patients on long-term therapy (de Vries 2020; Gadelha 2022).

Consider family planning, contraception, and the effects on fertility prior to prescribing opioids for chronic pain to patients who could become pregnant (ACOG 2017; CDC [Dowell 2022]).

Pregnancy Considerations

Fentanyl crosses the placenta (Leuschen 1990). Following epidural use for labor analgesia, fentanyl remained detectable in neonatal urine 24 hours after birth (Moore 2016).

Maternal use of opioids may be associated with poor fetal growth, stillbirth, and preterm delivery (CDC [Dowell 2022]). Opioids used as part of obstetric analgesia/anesthesia during labor and delivery may temporarily affect the fetal heart rate (ACOG 2019b). Transient muscular rigidity has been observed in the neonate following maternal administration of IV fentanyl; symptoms of respiratory or neurological depression were not different than those observed in infants of untreated mothers following IV or epidural use during labor.

Neonatal abstinence syndrome (NAS)/neonatal opioid withdrawal syndrome (NOWS) may occur following prolonged in utero exposure to opioids (CDC [Dowell 2022]). NAS/NOWS may be life-threatening if not recognized and treated and requires management according to protocols developed by neonatology experts. Presentation of symptoms varies by opioid characteristics (eg, immediate release, sustained release), time of last dose prior to delivery, drug metabolism (maternal, placental, and infant), net placental transfer, as well as other factors (AAP [Hudak 2012];

AAP [Patrick 2020]). Clinical signs characteristic of withdrawal following in utero opioid exposure include excessive crying or easily irritable, fragmented sleep (<2 to 3 hours after feeding), tremors, increased muscle tone, or GI dysfunction (hyperphagia, poor feeding, feeding intolerance, watery or loose stools) (Jilani 2022). NAS/NOWS occurs following chronic opioid exposure and would not be expected following the use of opioids at delivery (AAP [Patrick 2020]).

Monitor infants of mothers on long-term/chronic opioid therapy for symptoms of withdrawal. Symptom onset reflects the half-life of the opioid used. Monitor infants for at least 3 days following exposure to immediate-release opioids; monitor for at least 4 to 7 days following exposure to sustained-release opioids (AAP [Patrick 2020]; CDC [Dowell 2022]). Monitor newborns for excess sedation and respiratory depression when opioids are used during labor.

When opioids are needed to treat acute pain in pregnant patients, the lowest effective dose for only the expected duration of pain should be prescribed (CDC [Dowell 2022]).

Fentanyl injection is approved as an adjunct to general or regional anesthesia and as an analgesic during anesthesia. Pregnant patients should not be denied medically necessary surgery, regardless of trimester. If the procedure is elective, it should be delayed until after delivery (ACOG 2019a).

Fentanyl IM and IV injection are commonly used to treat maternal pain during labor and immediately postpartum; the intranasal route has also been studied (ACOG 2019b; Jabalameli 2016). Not all formulations are recommended. Opioid use for pain following vaginal or cesarean delivery should be made as part of a shared decision-making process. A stepwise, multimodal approach to managing postpartum pain is recommended. A low-dose, low-potency, short-acting opioid can be used to treat acute pain associated with delivery when needed (ACOG 2021).

Opioids are not preferred for the treatment of chronic noncancer pain during pregnancy; consider strategies to minimize or avoid opioid use. Advise pregnant patients requiring long-term opioid use of the risk of NAS/NOWS and provide appropriate treatment for the neonate after delivery. NAS/NOWS is an expected and treatable condition following chronic opioid use during pregnancy and should not be the only reason to avoid treating pain with an opioid in pregnant patients (ACOG 2017; CDC [Dowell 2022]). Do not abruptly discontinue opioids during pregnancy; taper prior to discontinuation when appropriate, considering the risks to the pregnant patient and fetus if maternal withdrawal occurs (CDC [Dowell 2022]).

Breastfeeding Considerations

Fentanyl and norfentanyl are present in breast milk (Cohen 2009).

The actual amount received by a breastfeeding infant varies. Reports are available following IV or epidural use during labor (Goma 2008; Leuschen 1990; Nitsun 2006; Steer 1992) or chronic maternal use of the transdermal patch (Cohen 2009). Not all studies evaluated concentrations of the active metabolite.

Nonopioid analgesics are preferred for lactating patients who require pain control peripartum or for surgery outside of the postpartum period. When opioids are needed for lactating patients, use the lowest effective dose for the shortest duration of time to limit adverse events in the mother and breastfeeding infant. Fentanyl may be used when an opioid is needed. Analgesics delivered by patient-controlled analgesia or administered by the epidural route help limit infant exposure (AAP [Sachs 2013]; ABM [Martin 2018]; ABM [Reece-Stremtan 2017]; WHO 2002).

When chronic opioids are prescribed prenatally and continued postpartum, breastfeeding may be initiated to help mitigate potential newborn withdrawal; monitor both the mother and the infant (AAP [Meek 2022]; AAP [Patrick 2020]).

Monitor infants exposed to opioids via breast milk for drowsiness, sedation, feeding difficulties, or limpness (ACOG 2019b). Withdrawal symptoms may occur when maternal use is discontinued or breastfeeding is stopped.

The Academy of Breastfeeding Medicine recommends postponing elective surgery until milk supply and breastfeeding are established. Milk should be expressed ahead of surgery when possible. In general, when the child is healthy and full term, breastfeeding may resume, or milk may be expressed

once the mother is awake and in recovery. For children who are at risk for apnea, hypotension, or hypotonia, milk may be saved for later use when the child is at lower risk (ABM [Reece-Stremtan 2017]).

Dietary Considerations

Transmucosal lozenge contains 2 g sugar per unit.

Monitoring Parameters

Pain relief; respiratory and mental status/alertness (especially in patients on concomitant CNS depressants, including benzodiazepines), BP, heart rate; signs of misuse, abuse, or substance use disorder; symptoms of serotonin syndrome such as mental status changes (eg, agitation, hallucinations, coma), autonomic instability (eg, tachycardia, labile BP, hyperthermia), neuromuscular changes (eg, hyperreflexia, incoordination), and/or GI symptoms (eg, nausea, vomiting, diarrhea); symptoms of neonatal withdrawal syndrome in babies born to mothers using opioids during pregnancy, including irritability, hyperactivity and abnormal sleep pattern, high-pitched cry, tremor, vomiting, diarrhea, and failure to gain weight; during discontinuation of therapy, monitor pain control, withdrawal symptoms, mood changes, suicidal ideation, and for use of other substances; signs or symptoms of hypogonadism or hypoadrenalism (Brennan 2013); signs or symptoms of opioid-induced esophageal dysfunction (eg, chest pain [noncardiac], dysphagia, regurgitation).

Transdermal patch: Monitor for 24 hours after application of first dose. Patients who experience adverse reactions should be monitored for at least 24 hours after removal of the patch. Drug continues to be absorbed from the skin for 24 hours or more following removal of the patch.

Alternate recommendations:

Subacute or chronic pain (long-term therapy outside of end-of-life or palliative care, active cancer treatment, sickle cell disease, or medication-based opioid use disorder treatment): Evaluate benefits/risks of opioid therapy within 1 to 4 weeks of treatment initiation and with dose increases. In patients with subacute pain initially treated for acute pain, reassess pain and function after 30 days to address potentially reversible causes of pain and prevent unintentional long-term opioid therapy. In patients on long-term therapy, re-evaluate benefits/risks every 3 months during therapy or more frequently in patients at increased risk of overdose or opioid use disorder. Toxicology testing is recommended prior to initiation and at least yearly (includes controlled prescription medications, illicit drugs of abuse, and benzodiazepines). State prescription drug monitoring program (PDMP) data should be reviewed by clinicians prior to initiation and periodically during therapy (frequency ranging from every prescription to every 3 months) (CDC [Dowell 2022]).

Critically ill: The Numeric Rating Scale should be used in patients who are able to self-report pain. In patients who are unable to self-report pain, the Behavioral Pain Scale and the Critical-Care Pain Observational Tool can be used in intubated or nonintubated patients (SCCM [Devlin 2018]).

Mechanism of Action

Binds with stereospecific receptors at many sites within the CNS, increases pain threshold, alters pain reception, inhibits ascending pain pathways

Pharmacokinetics (Adult Data Unless Noted)

Onset of action:

Children 3 to 12 years: Intranasal: 5 to 10 minutes (Borland 2002).

Adults: Analgesic: IM: 7 to 8 minutes; IV: Almost immediate (maximal analgesic and respiratory depressant effects may not be seen for several minutes); Transdermal patch (initial placement): 6 hours; Transmucosal: 5 to 15 minutes.

Duration: IM: 1 to 2 hours; IV: 0.5 to 1 hour; Transdermal (removal of patch/no replacement): Related to blood level; some effects may last 72 to 96 hours due to extended half-life and absorption from the skin, fentanyl concentrations decrease by ~50% in 20 to 27 hours; Transmucosal: Related to blood level; respiratory depressant effect may last longer than analgesic effect.

Absorption:

Transdermal, patch: Initial application: Drug is released at a nearly constant rate from the transdermal matrix system into the skin, where it accumulates; this results in a depot of fentanyl in the outer layer of skin. Fentanyl is absorbed into systemic circulation from the depot. This results in a gradual increase in serum concentration over the first 12 to 24 hours, followed by fairly constant concentrations for the remainder of the dosing interval. Absorption is decreased in cachectic patients (compared to normal size patients). Exposure to external heat increases drug absorption from patch.

Transmucosal, buccal tablet: Rapid, ~50% from the buccal mucosa; remaining 50% swallowed with saliva and slowly absorbed from GI tract.

Transmucosal, lozenge: Rapid, ~25% from the buccal mucosa; 75% swallowed with saliva and slowly absorbed from GI tract.

Distribution: Highly lipophilic, redistributes into muscle and fat; **Note:** IV fentanyl exhibits a 3-compartment distribution model. Changes in blood pH may alter ionization of fentanyl and affect its distribution between plasma and CNS.

V_{dss} :

Preterm and term neonates: **Note:** Volume of distribution is highly variable and significantly influenced by patient weight (Norman 2019; Wu 2022).

Bolus dose: 5.1 ± 1 L/kg (Koehtop 1986).

Infants, Children, and Adolescents ≤ 14 years of age (after long-term continuous infusion): ~15 L/kg (range: 5 to 30 L/kg) (Katz 1993).

Adults: 4 to 6 L/kg.

Protein binding: 79% to 87%, primarily to alpha-1 acid glycoprotein; also binds to albumin and erythrocytes; **Note:** Free fraction increases with acidosis.

Metabolism: Hepatic, primarily via CYP3A4 by N-dealkylation (to norfentanyl) and hydroxylation to other inactive metabolites.

Bioavailability: **Note:** Comparative studies have found the buccal tablet to have a 30% to 50% greater exposure than the transmucosal lozenge.

Buccal tablet: 65%.

Lozenge: ~50%.

Sublingual spray: 76%.

Half-life elimination:

IV:

Preterm and term neonate (PNA ≤ 14 days): Bolus dose: 5.28 ± 1.17 hours (Koehtop 1986); **Note:** Half-life may be prolonged (1.5 to 3 times the population mean) in patients with increased intra-abdominal pressure (Koehtop 1986).

Pediatric patients 5 months to 4.5 years: 2.4 hours.

Pediatric patients 6 months to 14 years (after long-term continuous infusion): ~21 hours (range: 11 to 36 hours).

Adults: 2 to 4 hours; when administered as a continuous infusion, the half-life prolongs with infusion duration due to the large volume of distribution (Sessler 2008).

SUBQ bolus injection: 10 hours (Capper 2010).

Transdermal patch: 20 to 27 hours (apparent half-life is influenced by continued fentanyl absorption from skin).

Transmucosal products: 3 to 14 hours (dose dependent).

Intranasal: 15 to 25 hours (based on a multiple-dose pharmacokinetic study when doses are administered in the same nostril and separated by a 1-, 2-, or 4-hour time lapse).

Buccal tablet: 100 to 200 mcg: 3 to 4 hours; 400 to 800 mcg: 11 to 12 hours.

Time to peak:

Buccal tablet: 20 to 240 minutes (median: 47 minutes).

Lozenge: 20 to 480 minutes (median: 20 to 40 minutes).

Intranasal: Median: 15 to 21 minutes.

SUBQ bolus injection: 10 to 30 minutes (median: 15 minutes) (Capper 2010).

Sublingual spray: 10 to 120 minutes (median: 90 minutes).

Transdermal patch: 20 to 72 hours; steady state serum concentrations are reached after two sequential 72-hour applications.

Excretion: Urine 75% (primarily as metabolites, <7% to 10% as unchanged drug); feces ~9%.

Clearance: Neonates: Clearance rapidly increased based on birthweight and PNA (Wu 2022); one study identified PNA as the most relevant covariate with very low clearance in the first few days after birth and then rapidly increasing until about 10 to 15 days of life (Völler 2019).

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: May alter kinetics because of alterations in clearance and plasma proteins.

Hepatic function impairment: May alter kinetics because of alterations in clearance and plasma proteins.

Pediatric: Plasma concentrations with transdermal use were approximately twice as high in pediatric patients 1.5 to 5 years of age, who are not opioid-tolerant, compared with adults. Pharmacokinetic parameters in older pediatric patients were similar to those seen in adults.

Older adult: Reduced clearance and terminal half-life is prolonged (transdermal).

Sex: Systemic exposure was higher in women after administration of buccal fentanyl (Fentora); this difference was largely attributed to differences in weight.

Race/ethnicity: Systemic exposure was higher in Japanese subjects after administration of buccal fentanyl (Fentora); this difference was largely attributed to differences in weight.

Extracorporeal membrane oxygenation (ECMO): Neonates, Infants, Children, and Adolescents: Increases in circulating blood volume and extraction of drug by the ECMO circuit (tubing and/or oxygenator) result in an increased volume of distribution (V_d) of drugs in patients on ECMO (Raffaelli 2019; Zimmerman 2020). As a result, larger doses/infusion rates may be required to achieve adequate pain control and level of sedation.

Medication Guide and/or Vaccine Information Statement (VIS)

An FDA-approved patient medication guide, which is available with the product information and as follows, must be dispensed with this medication:

Actiq: https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/020747s062lbl.pdf#page=44

Fentanyl transdermal patch: https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/019813s081lbl.pdf#p

Fentora: https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/021947s043lbl.pdf#page=43

Lazanda: https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/022569s030lbl.pdf#page=35

Onsolis: https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/022266s025lbl.pdf#page=32

Subsys: https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/202788s023lbl.pdf#page=40

Patient Counseling Points

(For additional information see "Fentanyl: Patient drug information")

What is this drug used for?

- It is used to manage pain.

All products other than injection:

- This drug is not for mild pain or pain that only lasts a short time (like headaches, toothaches, or pain after surgery).
- This drug is only for use by people who have been taking pain drugs (opioids) and are used to their effects. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

All products:

- Feeling dizzy, sleepy, tired, or weak
- Dry mouth
- Constipation, diarrhea, stomach pain, upset stomach, throwing up, or decreased appetite
- Headache
- Feeling cold
- Trouble sleeping
- Sweating a lot
- Irritation where this drug was used

Cheek tablet:

- Mouth tingling

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

All products:

- Low blood sugar like dizziness, headache, feeling sleepy, feeling weak, shaking, a fast heartbeat, confusion, hunger, or sweating
- Lowered interest in sex, fertility problems, no menstrual period, or ejaculation problems
- Severe dizziness or passing out
- Feeling confused
- A severe bowel problem like severe constipation or stomach pain
- Trouble breathing, slow breathing, or shallow breathing
- Noisy breathing
- Breathing problems during sleep (sleep apnea)
- Chest pain
- Fast or slow heartbeat
- Seizures
- Depression

- Trouble controlling body movements
- Swelling in the arms or leg
- Change in eyesight
- Serotonin syndrome like agitation; change in balance; confusion; hallucinations; fever; fast or abnormal heartbeat; flushing; muscle twitching or stiffness; seizures; shivering or shaking; sweating a lot; severe diarrhea, upset stomach, or throwing up; or severe headache
- Severe adrenal gland problem like if you feel very tired or weak, you pass out, or you have severe dizziness, very upset stomach, throwing up, or decreased appetite
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Cheek tablet:

- Mouth sores
- Swelling in the feet or hands

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Abstral | Fencino | Fentadur;
- (AR) Argentina: Durogesic d trans | Fentanilo celtyc | Fentanilo fabra | Fentanilo kilab | Fentora | Gray f | Lamuna | Sublimaze | Talnur;
- (AT) Austria: Actiq | Durogesic | Effentora | Fentabucc akut | Fentamed | Fentanyl | Fentanyl actavis | Fentanyl Genericon | Fentanyl hameln | Fentanyl janssen-cilag ph | Fentanyl sandoz | Fentanyl Stada | Fentanyl torrex pharma | Fentaplast | Fentoron | Instanyl | Ionsys | Lafene | Submena | Vellofent;
- (AU) Australia: Abstral | Actiq | Apo fentanyl | Denpax | Durogesic | Fenpatch | Fentanyl | Fentanyl sandoz;
- (BD) Bangladesh: Durogesic | Opifen;
- (BE) Belgium: Abstral | Durogesic | Effentora | Fentamax | Fentanyl matrix eg | Fentanyl matrix mylan | Fentanyl Matrix Sandoz | Instanyl | Matrifen;
- (BG) Bulgaria: Durogesic | Fentadolor | Fentanyl | Instanyl | Sublifen | Vellofent | Victanyl;
- (BR) Brazil: Anesfent | Biofent | Citrato de fentanila | Fentanest | Unifental;
- (CH) Switzerland: Abstral | Actiq | Durogesic | Effentora | Fentalis | Fentanyl actavis | Fentanyl Cimex TTS | Fentanyl helvepharm tts | Fentanyl Mepha | Fentanyl Sandoz MAT | Fentanyl sintetica | Fentanyl Spirig | Fentanyl spirig hc | Fentanyl zentiva;

- (CL) Chile: Durogesic | Fentadur | Fentanila | Fentora | Xalofenil;
- (CN) China: Durogesic d trans;
- (CO) Colombia: Diopal | Durfenta | Durogesic | Fentanex | Fentanilo | Fenyldol;
- (CR) Costa Rica: Durogesic | Fentanilo | Fentanyl;
- (CZ) Czech Republic: Adolor | Algogesic | Dolforin | Durogesic | Effentora | Fenroo | Fentalis | Fentalis akut | Fentanyl | Fentanyl ratiopharm | Instanyl | Lunaldin | Menasu | Pecfent | Vellofent;
- (DE) Germany: Abstral | Actiq | Durogesic | Durogesic Smat | Effentora | Fentaderm | Fentadolon | Fentamat | Fentanyl | Fentanyl 1 a pharma | Fentanyl AbZ | Fentanyl Actavis | Fentanyl AL | Fentanyl aristo | Fentanyl Awd Matrix | Fentanyl Beta | Fentanyl CT | Fentanyl Dura | Fentanyl Esparma | Fentanyl hameln | Fentanyl hennig | Fentanyl Heumann | Fentanyl hexal | Fentanyl Krewel | Fentanyl mylan | Fentanyl painbreak | Fentanyl puren | Fentanyl q pharm | Fentanyl Riemser TDS | Fentanyl tad | Fentanyl Winthrop | Fentapon | Fentavera | Fenyilat | Instanyl | Ionsys | Matrifen | Pecfent;
- (DK) Denmark: Instanyl;
- (DO) Dominican Republic: Durogesic | Fentanyl;
- (EC) Ecuador: Durfenta | Durogesic | Fentanex | Fentanyl;
- (EE) Estonia: Actiq | Durogesic | Effentora | Instanyl | Lunaldin | Matrifen;
- (EG) Egypt: Durogesic | Fentadur | Fentamat;
- (ES) Spain: Abfentiq | Abstral | Actiq | Avaric | Bimbax | Doloxital | Durfenta | Durogesic | Effentora | Fendivia | Fentanilo Actavis | Fentanilo Aristo | Fentanilo Cinfa | Fentanilo kalceks | Fentanilo Kern Pharma | Fentanilo matrix acino | Fentanilo matrix mylan | Fentanilo Matrix Normon | Fentanilo Matrix Sandoz | Fentanilo Stada | Fenticerta | Instanyl | Kaptic | Matrifen | Pecfent;
- (ET) Ethiopia: Fentanyl;
- (FI) Finland: Abstral | Actiq | Durogesic | Effentora | Fentanyl | Fentanyl actavis | Fentanyl Al-
pharma | Fentanyl braun | Fentanyl hameln | Fentanyl orion | Fentanyl ratiopharm | Fentanyl
sandoz | Fentanyl Stada | Instanyl | Matrifen | Pecfent;
- (FR) France: Abstral | Actiq | Durogesic | Effentora | Fentanyl actavis | Fentanyl arrow | Fentanyl
Biogaran | Fentanyl biogaran | Fentanyl dakota | Fentanyl EG | Fentanyl janssen | Fentanyl
lavipharm hellas | Fentanyl merck | Fentanyl mylan | Fentanyl Nycomed | Fentanyl ranbaxy |
Fentanyl ratiopharm | Fentanyl sandoz | Fentanyl Teva | Fentanyl Winthrop | Instanyl | Matrifen
| Pecfent | Recivit;
- (GB) United Kingdom: Abstral | Actiq | Cynril | Durogesic d trans | Fencino | Fenhuma | Fentalis
Reservoir | Fentanyl | Instanyl | Ionsys | Iremia | Mezolar | Mylafent | Opiodur | Osmanil | Pecfent
| Recivit | Victanyl | Yemex;
- (GR) Greece: Abstral | Actiq | Demogyl | Dolfen | Durogesic | Fentadur | Fentanyl orion | Fentanyl
sandoz | Fentanyl/mylan | Instanyl | Matrifen | Myfene | Pecfent | Vellofent;
- (GT) Guatemala: Durogesic | Fentanex;
- (HK) Hong Kong: Durogesic | Fentora;
- (HR) Croatia: Durogesic | Fentagesic | Fentanil kalceks | Fentanil M | Fentanyl | Matrifen;
- (HU) Hungary: Dolforin | Durogesic | Fentanyl | Fentanyl Pliva | Fentanyl ratiopharm | Instanyl |
Lunaldin | Matrifen;
- (ID) Indonesia: Durogesic;
- (IE) Ireland: Abstral | Actiq | Durogesic | Effentora | Fentadur | Fental | Instanyl | Matrifen |
Pecfent;
- (IL) Israel: Actiq | Beatryl | Durogesic | Fenta | Fentanyl;
- (IN) India: Durogesic | Fenstud | Finrid;
- (IT) Italy: Abstral | Actiq | Alghedon | Dogetic | Durfenta | Durogesic | Effentora | Fenroo | Fental-
gon | Fentanest | Fentanil ethypharm | Fentanil kalceks | Fentanil Ratiopharm | Fentanil sandoz |
Fentanyl | Fentanyl Lavipharm | Fenticer | Fenvel | Ifyltan | Instanyl | Pecfent | Sublifen | Vellofent;
- (JP) Japan: Abstral | Durotep | Durotep dainippon | Durotep mt | Durotep sankyo | Durotep sh-
ionogi | Durotep takeda | Durotep tanabe | E fen | Fentanest sankyo | Fentanest shionogi | Fen-
tanest tanabe | Fentanyl | Fentanyl 3 day | Fentanyl 3 day hmt | Fentanyl citrate tape for 1 day
daiichi sankyo | Fentanyl citrate tapes for 1 day | Fentanyl tape for 1 day meiji | Fentanyl tape
for 3 days towa | Fentanyl tapes for 1 day | Fentanyl tapes for 1 day meiji | Fentanyl tapes for 3
days | Fentos | Lafenta | Oneduro;
- (KE) Kenya: Durogesic | Fenstud | Fentavera;

- (KR) Korea, Republic of: Abstral | Actiq | Durogesic d trans | Fentaderm | Fentadur | Fentakhan | Fentamax Mat | Fentarix | Fentas | Fentavera | Fentora | Instanyl | Matrifen | Myungmoon fentanyl | Narco | Pecfent;
- (KW) Kuwait: Abstral | Durogesic;
- (LB) Lebanon: Abstral | Durogesic | Pecfent;
- (LT) Lithuania: Dolforin | Durogesic | Effentora | Instanyl | Lunaldin | Matrifen;
- (LU) Luxembourg: Durogesic | Fentanyl janssen | Fentanyl matrix eg | Instanyl | Matrifen;
- (LV) Latvia: Actiq | Dolforin | Durogesic | Effentora | Fentanyl | Instanyl | Lunaldin | Matrifen | Phentanyl;
- (MA) Morocco: Abstral | Durogesic | Fentanyl;
- (MT) Malta: Instanyl;
- (MX) Mexico: Durogesic d trans | Fentanest | Filtaten;
- (MY) Malaysia: Abstral | Durogesic | Fen touch | Fentanyl hexal;
- (NL) Netherlands: Abstral | Actiq | Durogesic | Effentora | Fentanyl | Fentanyl actavis | Fentanyl Apotex | Fentanyl aurobindo | Fentanyl glenmark | Fentanyl matrix | Fentanyl pch | Fentanyl ratiopharm | Fentanyl sandoz | Instanyl | Ionsys | Pecfent | Recivit;
- (NO) Norway: Abstral | Actiq | Actiq teva | Cynril | Durogesic | Effentora | Fentanyl ratiopharm | Fentanyl sandoz | Instanyl | Ionsys | Leptanal | Pecfent;
- (NZ) New Zealand: Biomed Fentanyl Citrat | Durogesic | Fentalis | Fentanyl | Fentanyl sandoz | Sublimaze;
- (PA) Panama: Durogesic;
- (PE) Peru: Durogesic d trans;
- (PH) Philippines: Abstral | Durogesic | Fendermal | Fentyn;
- (PK) Pakistan: Fentanyl sandoz | Fentavera;
- (PL) Poland: Aurofena | Dolforin | Durogesic | Effentora | Fenta MX | Fentagesic | Fentanyl | Fentanyl actavis | Fentanyl ratiopharm | Fentanyl Torrex | Instanyl | Lunaldin | Matrifen | Pecfent | Submena | Vellofent;
- (PR) Puerto Rico: Abstral | Actiq | Duragesic | Fentanyl | Fentanyl Cit | Fentora | Lazanaa | Subsys;
- (PT) Portugal: Abstral | Actiq | Durogesic | Fentanilo | Fentanilo ardicat | Fentanilo Ciclum | Fentanilo Matrifen | Fentanilo Mylan | Fentanilo sandoz | Fentanilo Wynn | Fentanilo zentiva | Instanyl | Nilfene | Pecfent | Vellofent;
- (PY) Paraguay: Durogesic | Fenium | Fentanilo interlabo;
- (QA) Qatar: Abstral | Durogesic | Matrifen | Sublimaze | Talinat;
- (RO) Romania: Algogesic | Dolforin | Durogesic | Effentora | Fentanyl actavis | Fentanyl TTS Sandoz | Instanyl | Lunaldin | Tabfyl;
- (RU) Russian Federation: Dolforin | Durogesic | Fendivia | Fentadol matrix | Fentanyl | Instanyl | Lunaldin;
- (SA) Saudi Arabia: Abstral | Durogesic | Fencino | Fentanyl | Pecfent;
- (SE) Sweden: Abstral | Actiq | Durogesic | Effentora | Fentanyl 2care4 | Fentanyl actavis | Fentanyl Alpharma | Fentanyl B Braun | Fentanyl hameln | Fentanyl hexal | Fentanyl Lavipharm | Fentanyl mylan | Fentanyl Orifarm | Fentanyl orion | Fentanyl ratiopharm | Fentanyl sandoz | Instanyl | Leptanal | Matriban | Matrifen | Pecfent | Submena;
- (SG) Singapore: Abstral | Durogesic | Fantamax | Fentanyl;
- (SI) Slovenia: Abstral | Durogesic | Effentora | Epufen | Fentanyl | Instanyl | Matrifen | Pecfent | Victanyl;
- (SK) Slovakia: Dolforin | Durogesic | Effentora | Fentanyl | Fentanyl actavis | Instanyl | Lunaldin | Submena | Vellofent;
- (TH) Thailand: Durogesic | Fentanyl GPO | Fentanyl hexal;
- (TN) Tunisia: Durogesic | Fentanyl;
- (TR) Turkey: Abstral | Durogesic | Effentora | Fentanyl | Fentanyl mercury;
- (TW) Taiwan: Durogesic | Fentanyl fresenius | Fentanyl Hpq | Fentora;
- (UA) Ukraine: Durogesic | Fentanyl | Fentanyl m | Fentavera | Matrifen;
- (UG) Uganda: Fentanyl sandoz;
- (UY) Uruguay: Durogesic | Fentanilo | Fentivo;
- (VE) Venezuela, Bolivarian Republic of: Durogesic | Fentanilo;
- (ZA) South Africa: Adco tenyl | Durogesic | Fendermal

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